A Summer Internship Project Report On

Industry Report on Wires and Cables at Nirmal Bang, Nashik

An internship report given with partial fulfillment of the requirement for Master of Business Administration (Finance) August 2025

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STUDENT DECLARATION

I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person not material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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3. Organizational Heads: Mr. Dilip Bang and Kishore Bang

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I Thank all of you for your guidance, support and invaluable contributions that led to the successful completion of this internship project

INTERNSHIP CERTIFICATE

CERTIFICATE OF FACULTY MENTOR

This is to certify that the summer internship report entitled **"Industry Report on Wires and Cables"** submitted by Shanu Singh to UPES for partial fulfilment of requirements for Masters of Business Administration (Marketing Management) is a Bonafide record of the internship work carried out by her/him under my (our) supervision and guidance. The content of the report, in full or parts have not been submitted to any other Institute or University for the award of any other degree or diploma.

Signature of Faculty Mentor

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CHAPTER: 1 INTRODUCTION

1.1 Abstract

This report offers an in-depth look at the Indian Wires and Cables (W&C) industry. It was prepared during a summer internship at Nirmal Bang, Nashik. The study is set against India's strong economic backdrop, which is a key driver of growth in this sector. By 2025, India is expected to be the world's fourth-largest economy by nominal GDP, approximately US$4 trillion, and is projected to reach US$5 trillion by FY26. The country has a consistent GDP growth rate of about 6.5%. This growth is supported by important factors, such as a young population with a median age of 28.8 years, rapid urbanization expected to hit 40% by 2030, a growing middle class, and rising digitalization. Together, these trends have increased private consumption, which grew at a CAGR of 10.4% from FY17 to FY22, making the consumer electricals market more attractive.

The Indian consumer electrical industry is valued at around ₹1.81 trillion in FY23, making it a key part of the national economy. Within this, the W&C sector holds a significant 41% share, with a market size of ₹74,800 crore in FY23. The industry is projected to grow at a strong CAGR of 13%, reaching ₹1.2 trillion by FY27. A major trend influencing the market is the clear move from unorganized to branded players. The market share of branded companies increased from 60% in FY15 to 76% in FY23. This change is driven by growing consumer awareness about safety and quality, along with the introduction of the Goods and Services Tax (GST), which has improved transparency and compliance across the industry.

1.2 Key Growth Drivers and Sectoral Demand

The growth of the W&C industry is driven by several strong factors from different sectors:

Infrastructure Investment: Large amounts of public and private funding are central to this growth. Important government programs like the National Infrastructure Pipeline (NIP), which plans an investment of ₹160 lakh crore, and the PM Gati Shakti National Master Plan are creating ongoing demand. The power sector, which makes up nearly 30% of W&C usage, is being significantly updated through initiatives like the Revamped Distribution Sector Scheme (RDSS), which has a budget of ₹3.03 trillion.

Real Estate Expansion: The residential real estate market, boosted by urbanization and programs like Pradhan Mantri Awas Yojana (PMAY), is the main user of housing wires. This segment is the largest and fastest-growing, holding a 32.8% market share in FY23. The strong commercial real estate sector, expected to expand at a 13% annual growth rate, adds to this demand.

New Age Sectors: The shift to a green and digital economy is leading to new, high-growth areas.

Renewable Energy: India's aim of reaching 500 GW of renewable capacity by 2030 requires a lot of cabling for solar parks and wind farms.

Electric Vehicles (EVs): The target of electrifying 30% of the vehicle fleet by 2030, along with the need to build 2 million charging stations, is opening a new market for specialized, heavy-duty cables.

Data Centres & 5G: Data centre capacity is expected to rise from 1.4 GW to 9 GW by 2030, and the rollout of 5G requires extensive fibre installation on telecom towers, which are currently only 35% equipped. This drives the need for optical fibre cables.

Rural Electrification: Programs like Saubhagya have provided electricity to 2.9 crore households. This initiative has created a large new consumer base and increased the need for last-mile connectivity infrastructure.

1.3 Market Segmentation, Competition, and Financial Landscape

The W&C market is divided into housing wires, power cables (LT, HT, EHV), communication cables, control and instrumentation cables, and flexible and specialty cables. Housing wires make up the largest segment and are expected to grow at a CAGR of 14.1%. Technological innovation plays a crucial role for brand leaders. Developments such as Low Smoke Zero Halogen (LS0H), Unilay Core Technology (UCT), and E-beam cross-linking improve safety, durability, and performance.

The competitive landscape shows increasing consolidation. The organized sector, led by five major players, Polycab, KEI Industries, Havells, Finolex, and RR Kabel, holds about 60% to 62% of the branded market. A financial review from FY21 to FY23 indicates strong performance among these leaders. Polycab leads in revenue, generating ₹14,108 crore in FY23. RR Kabel has shown the fastest growth, boasting a revenue CAGR of 43.4%. Profitability, shown through EBITDA margins, is closely linked to managing volatile raw material prices (copper and aluminium), which account for 60% to 70% of the selling price. Companies like Polycab, with a 13.9% margin, and KEI, with a 10.6% margin, have demonstrated strong operational efficiency.

Globally, India has become a net exporter of wires and cables, with exports valued at US$2.0 billion in CY23. The "China+1" global sourcing strategy offers a significant opportunity for Indian manufacturers to grow their international presence, taking advantage of their technical compliance and government export incentives like the RoDTEP scheme.

1.4. Executive Summary

The Indian Wires and Cables (W&C) industry plays a vital role in the national economy. Its market size reached ₹74,800 crore in FY23, and it is expected to grow at a strong 13% CAGR to reach ₹1.2 trillion by FY27. This growth connects closely to India's positive economic outlook, marked by steady GDP growth, rapid urbanization, and favourable demographics. Demand drivers for the industry's expansion include substantial government spending on infrastructure through the National Infrastructure Pipeline (NIP), a strong real estate market, and the fast development of new sectors like renewable energy, electric vehicles (EVs), and digital infrastructure including 5G and data centres.

A key trend is the significant market consolidation. Organized, branded companies have increased their market share from 60% in FY15 to 76% in FY23. This change results from greater consumer awareness about safety and quality, stricter enforcement of BIS standards, and the formal impact of the GST regime. The competitive landscape features five major firms—Polycab, KEI Industries, Havells, Finolex, and RR Kabel—that together dominate the branded market. Financially, the industry shows strong revenue growth. Companies like RR Kabel have displayed exceptional CAGR, but profitability is affected by fluctuating raw material prices, which can account for up to 70% of sales. On the operational side, India has established itself as a net exporter of wires and cables. Exports were valued at US$2.0 billion in CY23, benefiting from the global "China+1" sourcing strategy.

CHAPTER: 2 Overview of Global and Indian Economy

2.1 Macroeconomic Overview - GDP and GDP Growth

India's economy is expected to grow rapidly and remain relevant globally in 2025. India's economy continues to show resilience and momentum, currently ranking fourth in terms of nominal GDP (roughly US$ 4 trillion) and third in terms of purchasing power parity (PPP). Driven by industrial expansion, infrastructure development, and rising domestic consumption, the country's GDP is expected to grow by 6.5% in FY 2024–2025, putting it on track to reach US$ 5 trillion by FY 2026. The expansion of key industries is supported by this macroeconomic strength, especially the wire and cable sector, which is directly related to India's advancements in manufacturing, power, and infrastructure.

Table 1:GDP at current prices (In US$ trillion) and GDP ranking of key global economies (calendar year ("CY"))

country	Rank In GDP 2022	Rank in GDP (PPP)	CY 2010	CY 2015	CY 2020	CY 2022	CY 2025P	CAGR (2010-2021)	CAGR (2021-2024)
USA	1	2	15.1	18.2	21.1	25.3	30.5	4.1%	5.6%
China	2	1	6.1	11.1	14.6	17.8	19.2	10.2%	6.0%
Japan	3	4	5.7	4.4	5.1	4.2	4.1	-1.4%	2.7%
Germany	4	5	3.3	3.3	3.8	4.1	4.7	2.2%	5.9%
India	5	3	1.6	2.1	2.6	3.3	4.1	11.6%	11.9%
UK	6	10	2.4	2.9	2.7	3.1	3.8	1.4%	5.5%

Source: World Bank Data, Clear tax

India's nominal GDP has grown at a compounded annual growth rate ("CAGR") of 11.7% between CY 2010 and CY 2021 and is expected to continue the trend by growing at a CAGR of 11.9% between CY 2021 and CY 2025

From Fiscal 2010 until Fiscal 2022, the Indian economy's growth rate has been more than twice as that of the world economy and it is expected to sustain this growth momentum in the long term. India's economy is still growing faster than those of other big countries. In FY 2022, the country's GDP grew by 7.6% per year, and in FY 2024, it grew by 6.5% per year. Projections show that this growth will continue at a rate of 6.5% per year through FY 2025. This growth is much higher than the global average of 2.7% and the growth rates of other big economies like China (5.0%), Japan (0.1%), Germany (-0.2%), and the USA (2.8%) in CY 2024.

India's strong growth is due to a number of good structural factors, such as a young and growing workforce, more people moving to cities, new technologies, better digital infrastructure, and government reforms that move the country forward. These long-term enablers are likely to speed up economic activity in important areas.

Because of this, India is expected to be one of the top three economies in the world by FY

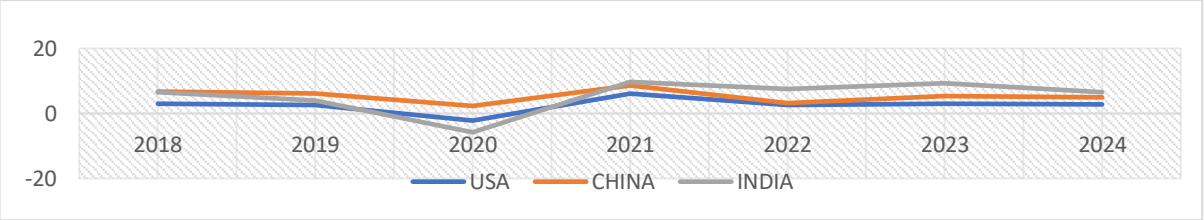


Figure 1

2030. This will provide a strong base for industries like wire and cable, which are important for building infrastructure and generating power.

Table 2:I GDP growth annual (%) – US, China, Japan, Germany, India (CY)

Annual GDP Growth	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025P
USA	2.9	1.8	2.5	3.0	2.6	-2.2	6.1	2.5	2.9	2.8	
China	7.0	6.8	6.9	6.8	6.1	2.3	8.6	3.1	5.4	5.0	
Japan	1.6	0.8	1.7	0.6	-0.4	-4.2	2.7	0.9	1.5	0.1	
Germany	1.7	2.3	2.7	1.1	1.0	-4.1	3.7	1.4	-0.3	-0.2	
India	8.0	8.3	6.8	6.5	3.9	-5.8	9.7	7.6	9.2	6.5	

Source: World Bank

2.2. Per Capita Income Growth

Income growth, presented by Gross National Income ("GNI") is defined as the total amount of money earned by a country's businesses and individuals

While India's gross national income grew at a CAGR of 8.1% for the period Fiscal 2015 to Fiscal 2022, GNI per capita for countries such as USA, China, Japan, Germany and UK grew at a CAGR of 3.5%, 6.9%, -0.6%, 0.9% and 0.0% between CY 2014 and CY 2021. Growing GNI is one of the strongest drivers for higher private consumption trends. The GNI per capita for the top five economies of the world for CY 2024 was USA (US$ 83,660), China (US$ 13,660), Japan (US$ 36,030), Germany (US$ 54,960) and the UK (US$ 48,610) as compared to India's GNI of US$ 2,650 (₹ 224,854) for a similar period of Fiscal 2024

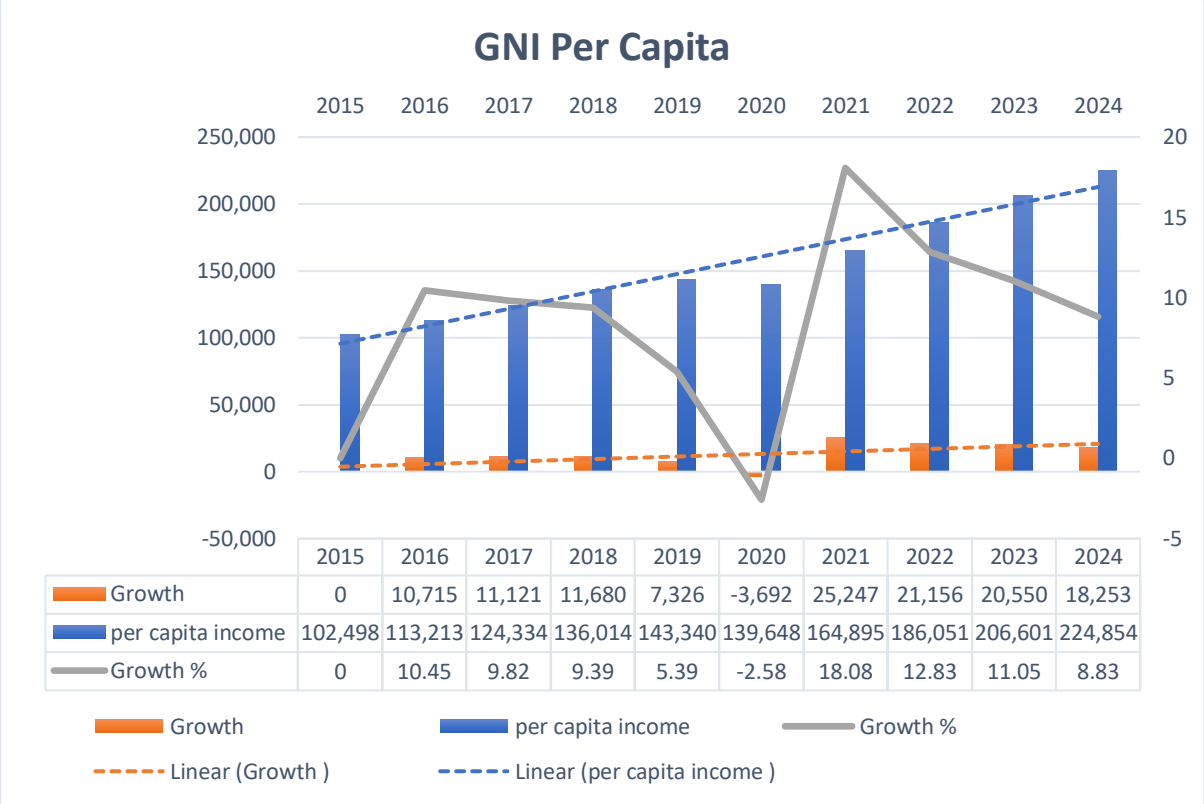


Figure 2:India's GNI Per Capita (₹) (Current Prices) and year-on-year ("Y-o-Y") growth trend in Fiscal Year

Source: World Bank group

2.3. Key characteristics of the Indian economy

2.3.1 Demographic profile of India

India has one of the youngest populations globally compared to other leading economies. The median age in India is estimated at 29.5 years for Fiscal 2024 as compared to 38.8 years and

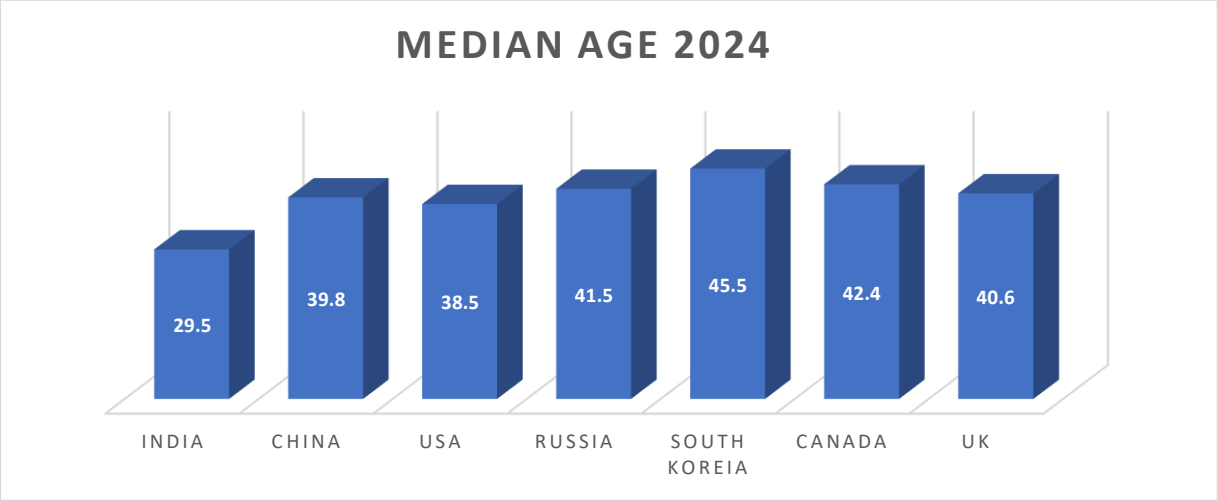


Figure 3

Source: World population Review

39.8 years in the United States and China, respectively, and is expected to remain under 30 years until Fiscal 2030. With a growing young population, the demand for premiumization is also growing. With more young people joining the workforce and earning higher salaries, there is a corresponding increase in disposable incomes, and hence the ability to afford premium products and services

India is still a demographic powerhouse; it has more people than China and is now the most populous country in the world. It is thought that by 2024, its population will be about 1.44 billion, which is about one-sixth of the world's population. One important thing about India's population is that it is big and young. The average age in the country is about 28.8 years, which is one of the lowest in the world. A large part of the population is of working age (15 to 64 years old), with more than 67% of the total population falling into this group as of 2024. This is a huge chance called the "demographic dividend." It means that there could be economic growth if there are more people of working age than people who depend on them (children and the elderly).

2.3.2. Private Consumption

Private Final Consumption Expenditure Growth (%)

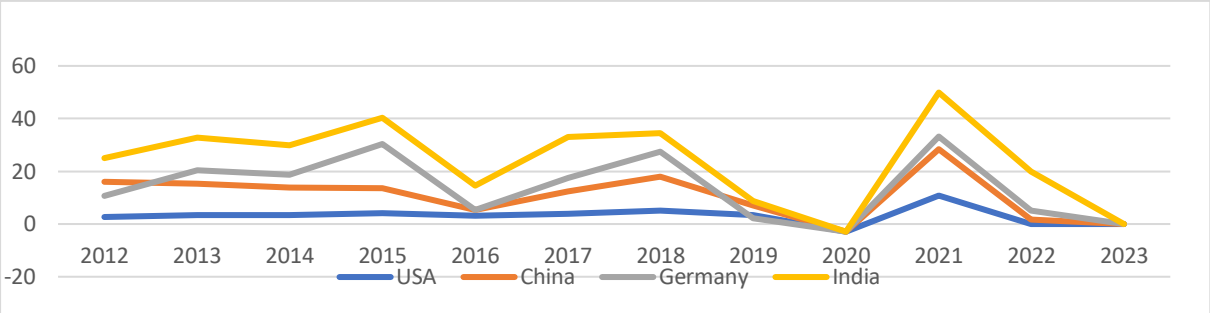


Figure 4:Private Final Consumption Expenditure Growth (%)

Country	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
USA	3.4%	4.1%	3.1%	3.8%	5.0%	3.5%	-2.9%	10.8%		
Germany	4.8%	-16.7%	0.0%	5.3%	9.5%	-5.0%	0.0%	4.8%	3.4%	
India	11.1%	10.0%	9.1%	15.4%	7.1%	6.7%	0.0%	16.7%	14.8%	
China	10.5%	9.5%	2.3%	8.5%	13.0%	3.6%	0.0%	17.6%	1.8%	

Table 3:Private Final Consumption Expenditure Growth (%)

Source: World Bank, RBI, Technopak Research & Analysis

DP growth in India is expected to be driven by rising private final consumption expenditure. India is a consumption driven economy where the share of domestic consumption is measured as private final consumption expenditure ("PFCE"). This private consumption expenditure comprises both goods (such as food, lifestyle, home and pharmacy) and services (such as food services, education and healthcare). A high share of private consumption to GDP has the advantage of insulating India from volatility in the global economy. It also implies that sustainable economic growth directly translates into sustained consumer demand for goods and services. India's domestic consumption has grown at a CAGR of 10.4% between Fiscal 2017 and Fiscal 2022, compared to 4.3% and 9.6% in the USA and China, respectively during the similar period of CY 2016 and CY 2021.

2.3.3. Urbanization

Urbanization is one of the most important pillars of India's growth story as these areas are the core drivers for consumption. India had the second largest urban population in the world in absolute terms at 522 million in CY 2025, second only to China. However, only 37% of India's population is classified as urban compared to a global average of approximately 57%. It is the pace of India's urbanization that is a key trend fuelling India's economic growth. Currently urban population contributes 63% of India's GDP. Going forward, it is estimated that 40% (550 million) of India's population will be living in urban centres by Fiscal 2030. This trend is expected to continue with approximately 50% of India's population expected to be living in urban centres by Fiscal 2050 respectively. India is the second largest urban system in the world with approximately 11% of the total global urban population living in its cities

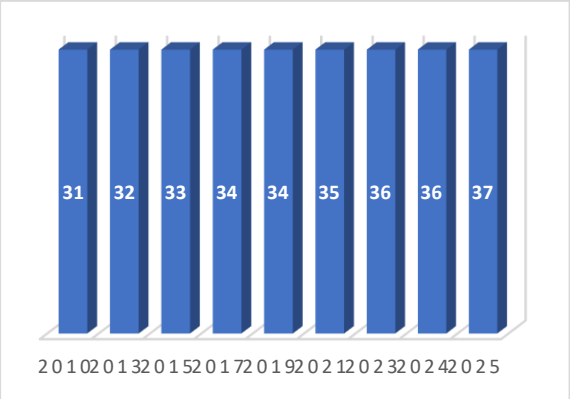


Figure 6:India's increasing Urban Population

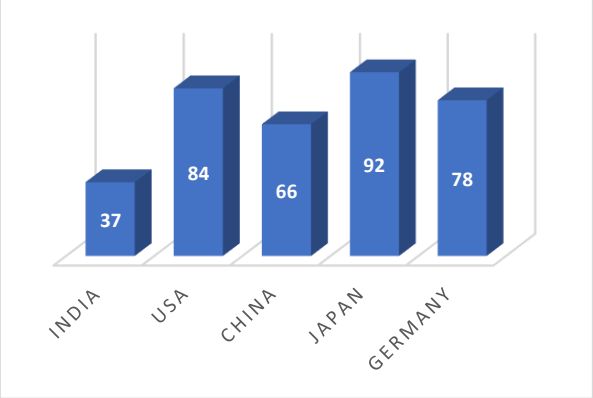


Figure 5:Urban Population as Percentage of Total

Source: World Bank group

2.3.4. Growing Middle Class

Increase in number of households with annual earnings of US$ 10,000 to US$ 50,000 is expected to drive the Indian economy due to demand of, among other things, more goods, better services, houses, health and education. Households with an annual income between US$ 10,000 and US$ 50,000 formed a minority stake of 5.8% of the total population in Fiscal 2010. This share increased to 30.6% in Fiscal 2020 and is expected to continue the growth momentum and increase to 42% of the total population by Fiscal 2030. Due to the growing middle-class

sector in India, there is an increasing appetite for premiumization of goods and services, construction, housing services, financial services, telecommunications and retail.

Household ("HH") Annual Earning Details

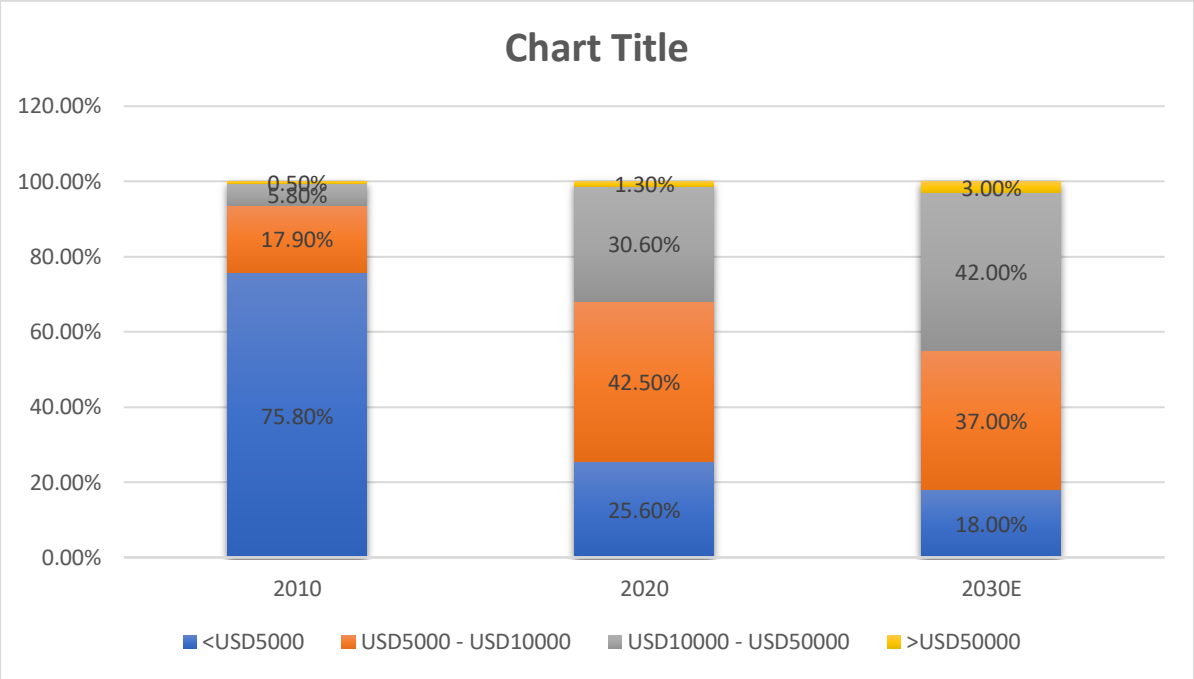


Figure 7:Household ("HH") Annual Earning Details

Source: RR Kabel DRHP

2.3.5. Nuclearization

The growth in the number of households exceeds the rate of population growth, which indicates an increase in nuclearization in India. The average household size has reduced from 5.3 in Fiscal 2001 to 4.2 in Fiscal 2021 and is further projected to reduce to 3.9 by Fiscal 2030. 69% of households had less than five members in Fiscal 2011 as compared to 62% in Fiscal 2001. Growth in the number of nuclear families is leading to an increase in the number of households, thereby creating a strong demand for housing units and discretionary expenditure in India.

Total number of households in India (In million)

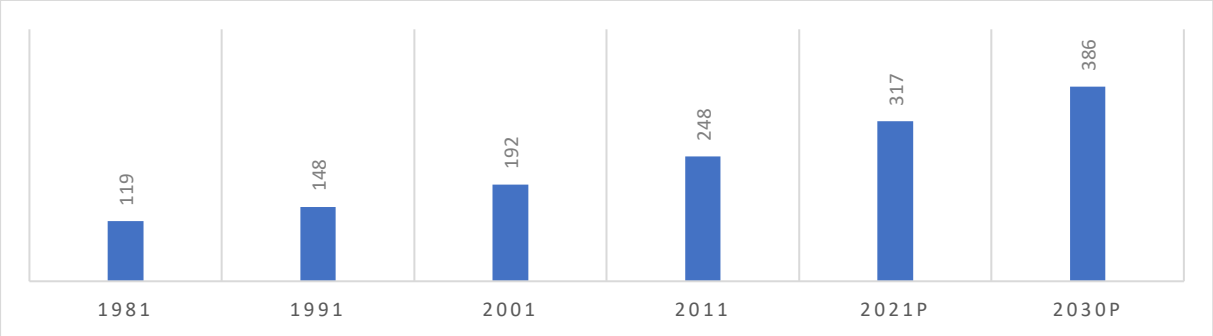


Figure 8:Total number of households in India (In million)

Source: RR Kabel DRHP

2.3.6. Digitization

Since 2010 India has undergone an extraordinary digital evolution, distinguished by an explosive rise in internet connectivity, a flood of smartphones, and a dramatic pivot to digital payments. From a fledgling start, internet coverage jumped from 7.5% to 52.4% by January 2024, totalling 886 million active users. Much of this growth has come from smartphones, whose ownership leaped from 76 million in 2013 to more than 926 million in 2024, confirming India's identity as a mobile-first digital economy

Growth of digital penetration in India (Fiscal Year)

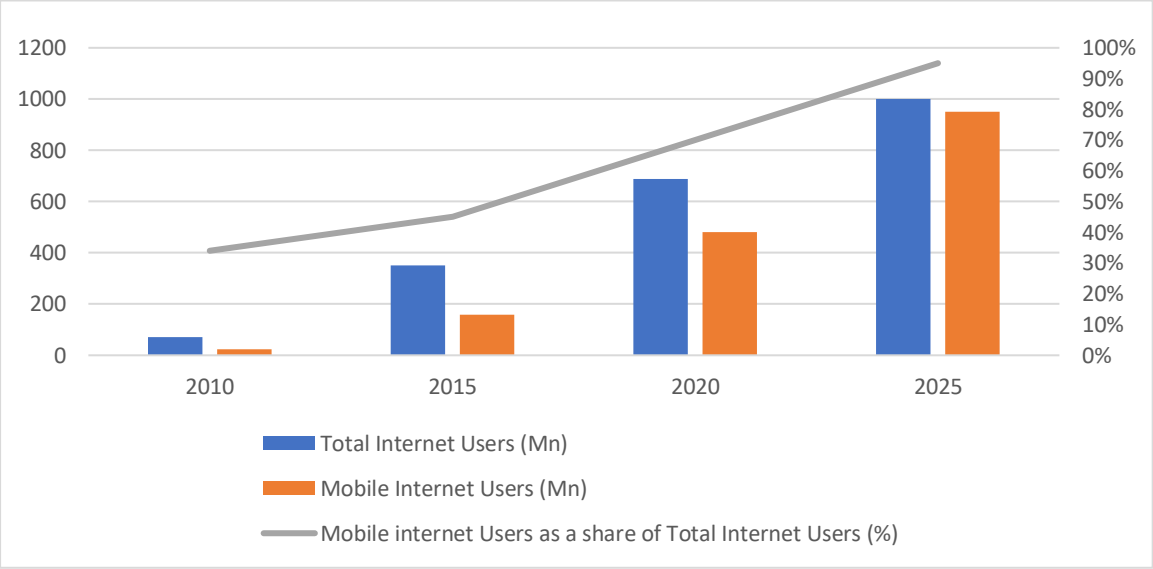


Figure 9:Growth of digital penetration in India (Fiscal Year)

Source: RR Kabel DRHP

The Unified Payments Interface (UPI) is a key part of this change. It processed over 16.73 billion transactions in December 2024, showing its strength in everyday use. Government programs like "Digital India" and the "Jan-Dhan-Aadhaar-Mobile" (JAM) trinity have played a crucial role in promoting financial inclusion and speeding up digital adoption, especially in rural areas where the number of internet users is growing faster than in cities. While we have reached notable milestones, challenges still exist in closing the digital gap, particularly regarding gender and rural access. However, with ongoing improvements in infrastructure, such as the fast rollout of 5G, and a young population eager for technology, India's digital economy is expected to keep growing, reinforcing its position as a global leader in the digital space.

Urbanization and nuclearization is expected to increase the demand for residential real estate in India, which in turn, is expected to generate demand for electrical products as each house would require, among other things, electrical appliances, wirings, fans and 72 350 687 778 788 1000 24 159 480 754 760 950 34% 45% 70% 97% 96% 95% 0% 25% 50% 75% 100% 0 300 600 900 1200 2010 2015 2020 2021 2022 2025 (P) Growth of digital penetration in India Total

Internet Users (Mn) Mobile Internet Users (Mn) Mobile internet Users as a share of Total Internet Users (%) Exhibit 1.9: Total number of households in India (In million) Source: Census, Technopak Analysis; Note: Decadal growth for period 2021-2030P reflects a 9-year period 75.8% 25.6% 18.0% 17.9% 42.5% 37.0% 5.8% 30.6% 42.0% 0.5% 1.3% 3.0% 2010 2020 2030P HHs with Annual earning greater than USD 50,000 HHs with Annual earning USD 10,000 –50,000 HHs with Annual earning USD 5,000- 10,000 HHs with Annual earning less than USD 5000 119 148 192 248 317 386 1981 1991 2001 2011 2021P 2030P lights. India's youth and working age population, accompanied by rise in per capita income is expected to drive premiumization in fast moving electrical goods ("FMEG") segments such as fans, lightings and home appliances. Additionally, the pace at which technology adoption and digitization is happening across India, due to factors such as improvement in telecommunications infrastructure and increase in data consumption, is expected to increase the demand for wires and cables ("W&C") such as communication cables in India. The Government of India ("GOI") has announced policies aimed at boosting GDP, increasing the share of manufacturing in India and economic growth, thereby boosting electricity consumption by the industrial segment in India. GOI has also announced various schemes to revive power infrastructure, improve urban distribution networks such as Integrated Power Development Scheme ("IPDS") and boost rural electrification. Government initiatives in housing and infrastructure sectors like Pradhan Mantri Awas Yojana ("PMAY") and PM Gati Shakti National Master Plan ("NMP") will likely increase the demand for electrical products in India. Construction of Multimodal logistics parks ("MMLPs") is expected to increase the demand for wires and cables such as flexible cables, control and instrumentation cables, and construction of newly built houses is expected to drive the demand for housing wires and other FMEG products

CHAPTER 3: Consumer Electrical Industry in India

3.1. Market Size and Segmentation

India occupies a vital position in the global electricity landscape, standing as the third-largest producer and the second-largest consumer of electric power. Its consumer electrical sector is a key pillar of the national economy, contributing around 8% of overall manufacturing output, about 1.5% of the gross domestic product, and the same proportion to national exports. The sector is wide-ranging, spanning heavy electrical and light electrical segments. The heavy side includes Wires and Cables, where “wires” refer to single conductors and “cables” denote bundled conductors, both designed to carry electricity as well as telecommunication signals. The light electrical segment, often called Fast Moving Electrical Goods, includes a broad assortment of end-user products such as ceiling and table fans, luminaires, electric heaters, and assorted household appliances.

The total market for consumer electrical goods, which includes both Wires and Cables (W&C) and Fans, Motors, and Electrical Goods (FMEG), was around ₹1,81,150 crore in FY 2023. Analysts expect the combined market to expand at about 10% CAGR and hit ₹2,66,500 crore by FY 2027. Within that picture, W&C holds a large piece, accounting for close to 40% of the overall W&C and FMEG sectors in India. The stand-alone W&C market was valued at ₹74,800 crore in FY 2023 and is forecast to grow at 13% CAGR, targeting ₹1,20,000 crore by FY 2027.

The FMEG division, though labelled as "light electrical products," commands a significant proportion of the entire consumer electrical sector. For Fiscal 2023, the Indian FMEG marketplace was valued at roughly ₹1,06,350 crore, making up roughly 59% of the complete consumer electrical sector. Analysts predict an 8% compound annual growth rate through Fiscal 2027, which would lift the segment to about ₹1,46,500 crore. Within the FMEG category, a closer examination shows home appliances represented 28% and kitchen appliances the remaining 72% of the Indian home and kitchen appliances segment during Fiscal 2023.

More recent external data gives insight into the larger consumer electronics market, which overlaps a lot with FMEG. The size of the India Consumer Electronics market reached USD 83.70 billion in 2024. It is expected to grow to USD 152.59 billion by 2033, showing a CAGR of 6.90% during this time. While "FMCG" (Fast Moving Consumer Goods) is a broader term, some reports include consumer electronics in this group. For example, the India FMCG market was valued at USD 245.39 billion in 2024. It is estimated to hit USD 1,108.48 billion by 2033, with a CAGR of 17.33%. Another source lists the FMCG market size at USD 199 billion in 2024. It is projected to reach USD 211 billion in 2025, with a CAGR of 21.8% from 2025 to 2034. These numbers, while covering a wider range of products, highlight the strong growth in consumer-driven sectors that directly affect the demand for electrical goods.

Rural electrification, expansion in the residential and commercial real estate sector, increase in per capita consumption of electricity, Government of India’s (“GoI”) infrastructure development support through increased capital outlay and government funded schemes, favourable trade policies to increase exports through initiatives such as production linked incentive (“PLI”) and Remission of Duties and Taxes on Exported Products (“RoDTEP”) schemes, successful execution of Bureau of Energy Efficiency (“BEE”) star norms and increase

in demand from consumers to upgrade or replace existing products are some of the key factors, that are expected to drive the growth of the consumer electrical industry in India.

Increase in exports of cables & wires higher than that of imports

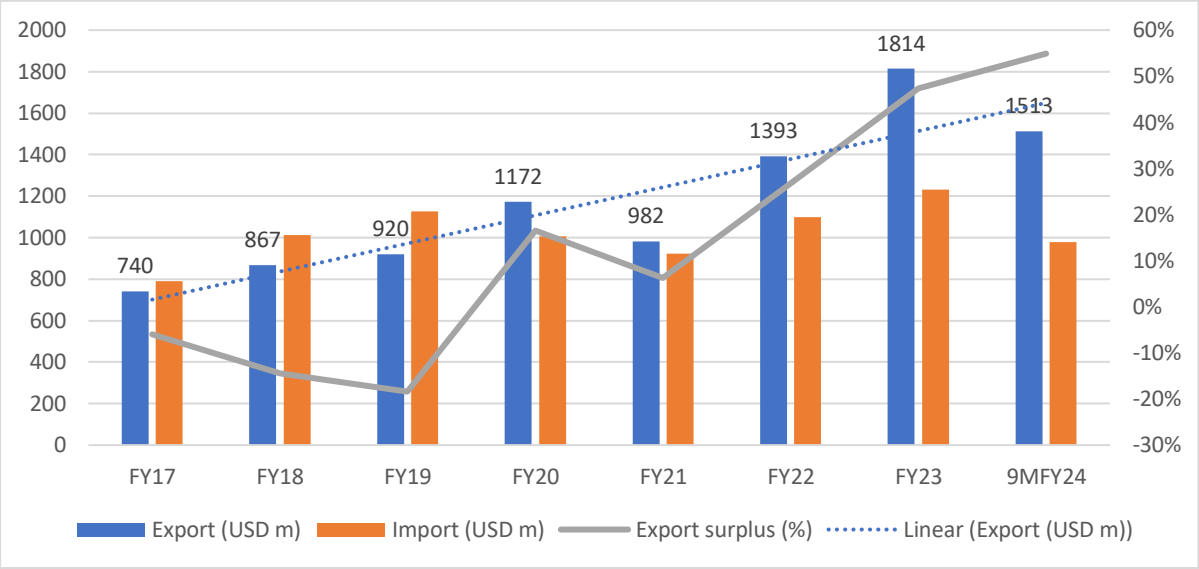


Figure 10:rease in exports of cables & wires higher than that of imports

Source: RR Kabel DRHP

Indian FMEG Market - Value (INR Cr)

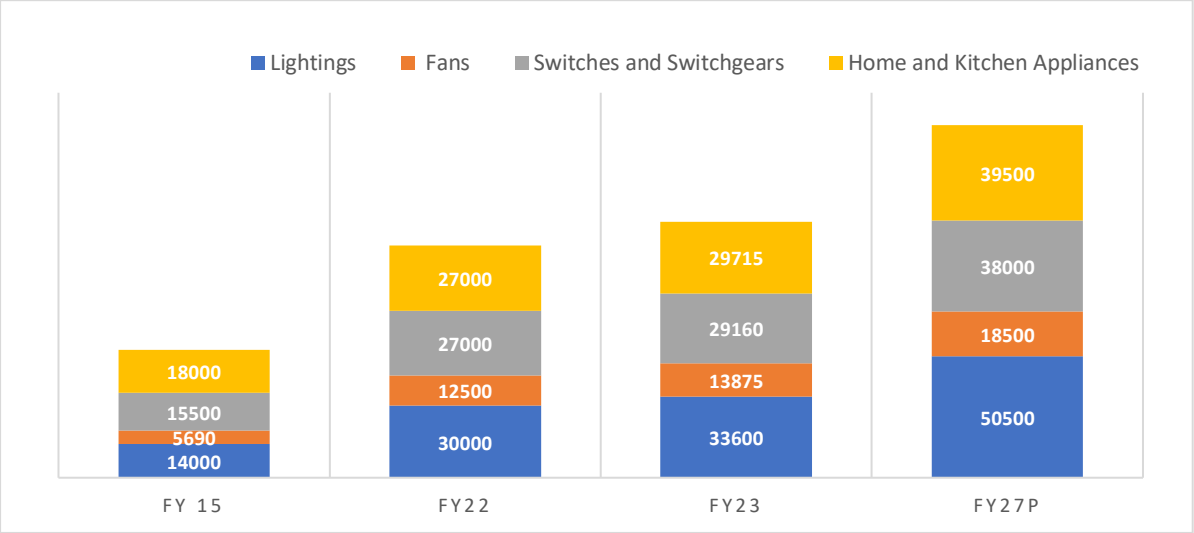


Figure 11:Indian FMEG Market - Value (INR Cr)

Source: RR Kabel DRHP

The Indian consumer electrical industry's market size stood at INR1.8t as of FY23. The industry recorded a CAGR of 9-10% over FY16-23. The industry growth has been fuelled by strong government focus on infrastructure development, rising trend of nuclear families, increase in electrification in rural areas, increase in exports of cables & wires, etc.

Market size of different segments of the Indian electrical industry (INR b)

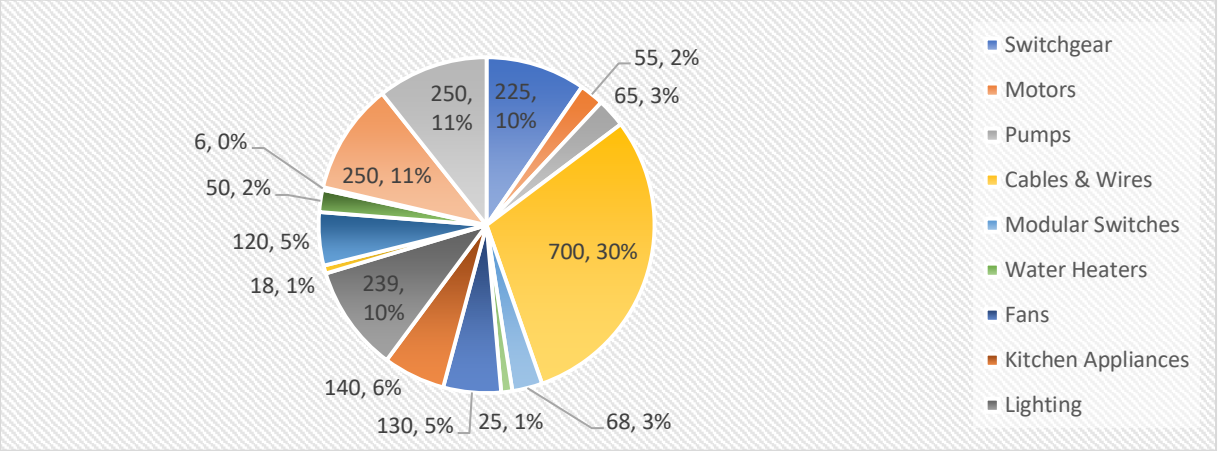


Figure 12:Market size of different segments of the Indian electrical industry (INR b)

Source: MOFSL

According to our estimates, the industry should clock ~10% CAGR over the next few years, with higher growth estimated for cables & wires (12-14% CAGR until FY27E), water heaters, fans, etc. The industry would also benefit from the government policies, e.g., encouraging domestic manufacturing and PLI incentives for components. There could also be a shift to organized players from unorganized ones, which has historically been the case. This possible shift would further benefit the bigger players going forward.

3.2. Trends shaping W&C and FMEG industry in India

3.2.1. Shift Towards Branded Play

A significant and growing trend in the Indian consumer electrical industry is the clear movement toward branded products. The share of branded companies in the domestic W&C and FMEG industry has risen from 60% in Fiscal 2015 to about 76% in Fiscal 2023. Projections suggest this will consolidate further to around 82% by Fiscal 2027. This important change is driven by both demand and supply factors.

3.3. Demand side factors

Rising Awareness among consumers towards safety and quality: There has been a growing awareness among the consumers to seek safety and quality in electrical products. Given products' interface involves electricity and electricals, there is a heightened consumer awareness about branded play being a proxy for trust, quality, and safety. Thus, customers across income segments have increasingly started to prefer branded products over unbranded products. This consumer behaviour has led the electrical products market to get pivoted towards branded play.

3.4 Supply side factors

On the supply side, the introduction of the Goods and Services Tax (GST) has had a major impact. GST has required more transparency throughout the entire value chain, from manufacturers to retailers. This has discouraged illegal trade practices like under-reporting

production and sales, non-billed transactions, and non-compliant behaviour. Additionally, the availability of input tax credit for taxes paid at various stages of the value chain has made trading branded products more financially viable. At the same time, increased GST compliance has raised costs for unbranded players. These changes have effectively closed the price gap between branded and unbranded products, creating a better environment for branded companies to grow their market share.

Another important supply-side factor is the continuous **technological innovation** brought by branded players. Innovations like Low-Smoke Zero-Halogen (LSZH) technology in wires and cables improve safety by lowering smoke and toxic gas emissions during fires. The ongoing premiumization trend in fast-moving electrical goods (FMEGs) has also led to significant product differentiation. These advancements are often hard for unbranded players to imitate because they require substantial investment in research, development, and manufacturing. As a result, the combination of consumer preference for safety and quality, regulatory changes that encourage transparency, and technological advantages has driven the market firmly towards branded products.

3.5. Key Characteristics of the Industry

The Indian consumer electrical industry exhibits several distinct characteristics that define its operational and competitive dynamics.

3.5.1. Direct Consumer Sales Activation

Historically, in the 1990s, about 80-90% of sales in the W&C and FMEG industry depended on distributors and retailers. This gave brands limited insight into sales to end consumers. The landscape has changed significantly. Currently, around 45% of sales come from a brand's direct sales team, which works with distributors and retailers. About 50% of sales still rely on distributors and retailers, while roughly 5% occur through a brand's exclusive stores. Companies are increasingly pushing for direct sales to customers on-site, including electricians, who are key decision-makers in this industry. This shift brings several benefits: it helps eliminate local or regional "trade labels" (unbranded products), reduces reliance on individual dealers, strengthens brand image through direct customer interaction, and provides important Below-the-Line (BTL) marketing support for Above-the-Line (ATL) investments. Additionally, the rise of sales through exclusive brand outlets (EBOs) improves direct consumer interaction, increases brand visibility, and supports the premiumization trend by offering a controlled retail experience.

3.5.2. Marketing Initiatives (ATL & BTL)

Brand building in the consumer electrical industry relies on two approaches: ATL and BTL marketing initiatives. ATL marketing is essential for creating brand presence and awareness. Companies use various media channels, including television, print, outdoor billboards, movie theatres, and digital platforms. Celebrity endorsements often play a key role in these campaigns. They typically aim to showcase product features like safety, durability, energy efficiency, design, and functionality. Leading companies such as RR Kabel invest a significant portion—close to 40%—of their marketing budget in ATL efforts, which include celebrity endorsements and sponsorships of popular sports teams.

To support ATL efforts, BTL marketing initiatives are important for boosting brand awareness and consumer recall at a more local and direct level. These activities include sales promotions, visual merchandising, direct marketing, leaflets, and countertop displays. Key players like RR Kabel dedicate about 60% of their marketing budget to BTL activities, such as loyalty programs for electricians (e.g., RR Connect App) and educational programs for architects and electrical consultants (e.g., Kabel Link). These programs aim to build a network of influencers, improving brand awareness and visibility directly at the point of sale or influence.

3.5.3. Combination of In-house and Outsourcing

Within the W&C and FMEG industry, manufacturers often use a hybrid strategy for product sourcing. They combine in-house manufacturing, imports, and outsourcing to third-party manufacturers. Companies like RR Kabel and KEI have integrated their W&C manufacturing facilities, producing some raw materials like PVC compounds in-house. This gives them more control and visibility over product quality. In contrast, players in the lighting industry mostly rely on importing finished goods. Those in the home and kitchen appliances sector use a mix of outsourced and in-house manufacturing. This flexible strategy helps companies cut costs, take advantage of specialized manufacturing skills, and keep quality standards high across their various product lines.

Estimated In-house vs outsourcing share for different players in Consumer Electrical Industry (Fiscal 2023)

Table 4:Estimated In-house vs outsourcing share for different players in Consumer Electrical Industry (Fiscal 2023)

Companies	In-house Manufacturing (%)	Outsourced (%)
RR Kabel	93%	7%
Havells	90%+	<10%
Polycab	95%	5%
KEI	95%	5%
Finolex	97%	3%
V-Guard	45%	55%
Crompton	NA	NA
Bajaj Electricals	NA	NA

Source: RR Kabel DRHP

3.6. Key Growth Drivers of Consumer Electrical Industry in India

There are six key factors supported by a positive macro environment that are expected to provide growth in the W&C and FMEG industry between Fiscals 2025 and 2032:

1.Public and private investment outlay in infrastructure

2. Continued growth of residential real estate sector

3.Resilient commercial real estate sector

4. Transition of automobiles and transport towards electric vehicles ("EVs")

5. Rural electrification

6. Push towards renewable energy

3.6.1. Public and private investment outlay in infrastructure

Stimulus alongside wires and cables still increase due to the Indian government's pending infrastructure further development. Policy frameworks and initiatives of the PM GatiShakti National Master Plan seeks to foster and improve multi-modal connectivity of the whole country. Also, the National Infrastructure Pipeline (NIP) projected an investment of ₹111 lakh crore (approximately US$1.5 trillion) from FY2020-2025. Moreover, the Revamped Distribution Sector Scheme (RDSS) has been designed towards the annual spend of ₹3.03 trillion (USD 35.8 billion) from 2021-22 to 2025-26 and aims exclusively at updating and modernizing the country's electricity distribution infrastructure.

Power Generation & Transmission:

Substantial funds are being set aside for new power capacity additions, expected to be between 45 and 55 GW, along with the construction of 10,000 km of new inter-state transmission lines. India's installed power generation capacity is projected to rise from 517 GW in 2023 to 781 GW by 2028. This growth will require a strong transmission and distribution network. The power utility sector makes up nearly 30% of the total wire and cable usage in India.

Roads & Highways:

The National Highways network saw major growth, increasing from 91,300 km in 2014 to 146,000 km in 2024. The rate of highway construction has nearly tripled in the past decade, rising from 12 km per day to 34 km per day. Key programs like Bharatmala Pariyojana have finished over 18,900 km of highways.

Railways:

India's vast railway network is experiencing its biggest upgrade in decades, backed by a budget of $2.7 billion for FY2023-24. This amount is nine times what was allocated in 2013-14. This investment helps fund important projects like metro expansions and railway electrification.

3.6.2. National Infrastructure Pipeline (NIP)

The National Infrastructure Pipeline (NIP) is a cornerstone of India's long-term growth strategy, and it's playing a major role in driving demand in the wire and cable industry.

Launched with a vision to transform India's infrastructure landscape, the NIP originally outlined investments worth ₹111 lakh crore (approximately $1.5 trillion) for the period FY2020 to FY2025. Since then, the scale of ambition has grown substantially, with the total projected investment rising by around 40–45% to ₹160 lakh crore. This sustained, multi-year commitment ensures a stable and growing demand for wires and cables across the country, especially as infrastructure projects gather pace.

Around 71% of the NIP's total investment is targeted at four major sectors: energy, roads, urban infrastructure, and railways — all of which are heavily dependent on cable infrastructure.

In the energy sector alone, nearly half the funds are allocated to renewable energy projects, followed by about 27% for conventional power and the remaining 25% for distribution, transmission, and storage infrastructure. This breakdown is particularly significant for the cable industry because transmission and distribution require high-performance, high-voltage, and often specialized cables.

So, whether it's laying power lines for solar parks, electrifying railways, expanding metro networks, or developing smart cities, cables are a core enabler. The NIP is not just about concrete and steel; it's about connectivity — electrical, digital, and transport — all of which rely on an efficient and scalable cable network.

Capex by Go

Capital investment of approximately ₹ 10 lakh crore on infrastructure sector (approximately 3.3% of GDP) has been announced by GoI in the Indian union budget 2023-24. Additionally, a capital outlay of ₹ 2.4 lakh crore has been provided for railways. The principal areas identified for capital expenditures include roads, railways and metros, which require high supplies of W&C and FMEG products. With the expansion of transport infrastructure, Technopak expects there to be a corresponding increase in demand for electrical goods such as W&C, lighting and switchgears, thereby creating a market opportunity for W&C and FMEG manufacturers

Pradhan Mantri Awas Yojana (PMAY)

The GoI under the 'Housing for All' scheme, has aimed to build approximately 12 million houses for urban and approximately 28.5 million houses in the rural parts of India. The Pradhan Mantri Awas Yojana (PMAY) scheme is important for creating demand for wires and cables, especially in residential areas. Scheme Allocation and Impact on Low Voltage (LV) Cable Demand: India's Finance Ministry has set aside over $9.5 billion for the PMAY scheme, showing the government's strong commitment to affordable housing and infrastructure. This initiative has greatly increased the demand

for Low Voltage Energy (LVE) cables. LVE cables now hold the largest market share, accounting for 34.7% of the total cable demand in India. Projections show that LVE cable consumption is expected to grow by 9.5% year-on-year in 2024. These initiatives are expected to generate demand for W&C such as housing wires, power cables, flexible and specialty cables and for FMEG products like lights, fans, home, and kitchen appliances in the next 5-7 years.

Urbanization and Smart Cities Mission

In the last decade, the urban population in India has grown at an annual rate of 2.4%. The urban population of India is projected to reach approximately 41% by CY 2030 from 31% in CY 2010. Further, the number of metropolitan cities in India are estimated to increase from 46 as per Census 2011 to 68 in CY 2030. Additionally, Smart City project was launched in June 2015 by GoI with a plan to build 100 smart cities across India, having an objective to promote cities that provide core infrastructure, good quality of life to its citizens, a clean and sustainable environment with application of 'Smart' solutions. As on December 2, 2022, GOI has released approximately ₹ 34,675 crore as part of Smart City project, of which approximately ₹ 30,418 crore (88%) have been utilized in building smart cities across India. These initiatives are expected to generate demand for electrical products especially in the W&C, switches and switch gear segments in next 5-7 years

Demand for Advanced Cabling: This urbanization trend drives a significant demand for advanced cabling solutions, including: Underground cabling projects in major cities like Delhi and Mumbai focus on moving away from traditional overhead cables. This shift aims to cut down power outages, increase reliability, and improve the look of urban areas. The rollout of smart grids depends on IoT-enabled cables. These cables help manage demand responses and allow for easy integration of rooftop solar systems. Strong cable networks are necessary for smooth energy distribution, effective communication systems, efficient transportation, security and surveillance, smart homes, and combining renewable energy sources. Fiber optic cables play a vital role in providing high-speed internet, smooth IoT connectivity, and real-time data transfer. These elements are key for automated city management systems, AI-driven analytics, and smart traffic control. Heavy-duty cables and energy-efficient conductors are also crucial for the growing electric vehicle (EV) charging infrastructure in smart cities.

Role of Revamped Distribution Sector Scheme (RDSS):

The Revamped Distribution Sector Scheme (RDSS) has a large budget of ₹3.03 lakh crore for the period from 2021-22 to 2025-26. It aims to improve cable systems. Its main goal is to cut Aggregate Technical & Commercial (AT&C) losses to below 12-15% by 2025. This scheme is crucial for updating India's electricity transmission and distribution infrastructure. Urbanization and Smart Cities initiatives are not just about increasing cable volume. They are changing the types of cables needed. The move toward underground cabling, IoT-enabled smart grids, and high-speed fibre optics shows a strong need for high-quality, high-performance, and specialized cabling solutions. These solutions support complex system integration and improve overall

efficiency. RDSS strengthens this trend by promoting upgrades aimed at reducing losses, which directly helps manufacturers of high-quality cables that can support smart infrastructure.

Bharat net

The Bharat Net project is a game-changer for India's digital infrastructure, significantly boosting demand for optical fibre cables. With a massive ₹1.39 lakh crore investment in Phase 3, the initiative aims to connect 6 lakh villages and 2.5 lakh gram panchayats to high-speed broadband over the next three years. It also includes fibre connectivity to mobile towers, setting the stage for advanced technologies like 5G and 6G. As India targets fibre broadband for 120 million homes and hyperscale data centres continue to expand, demand for optical fibre is expected to grow at a strong 12.5% CAGR through 2030. In fact, domestic fibre production crossed 100 million fibre-km in FY24, turning India into a net exporter. Bharat Net's scale and focus on rural connectivity mean high-volume and specialized cable demand, such as bend-insensitive fibres for tough terrains. More than just improving access, the project is laying a future-ready digital backbone for the country.

3.6.3. Policy Reforms

PLI Scheme:

Under PLI scheme, GoI aims to have large-scale electronics manufacturing in India. ₹ 1.97 lakh crore have been allocated for PLI in Indian union budget 2020-21 for next 5 years, which is expected to boost production by over $500 billion by FY 26. There is a PLI Scheme specifically for white goods (air conditioners and light emitting diode ("LED") lights) manufacturers in India under which, GoI proposes a financial incentive to boost domestic manufacturing and attract large investments in white goods manufacturing value chain, with an objective of removing sectoral disabilities, creating economies of scale, enhancing exports, creating a robust component ecosystem and employment generation. This scheme is expected to be implemented over Fiscal 2022 to Fiscal 2029 with a budgetary outlay of ₹ 6,238 crore. A total of 57 firms with committed investment of approximately ₹ 6,000 crore have been provisionally selected as beneficiaries under this scheme.

Integrated Power Development Scheme ("IPDS"):

IPDS envisages strengthening of sub-transmission and distribution network including metering at all levels in urban area. Till March 2023, projects worth ₹ 31,314 crore have been sanctioned under IPDS, against which, ₹ 15,922 crore have been released towards projects and ₹ 219 crore released for enabling activities.

Export reforms:

Various conducive export policies have been announced by GoI. For example, RoDTEP scheme has been introduced which provides rebates to various central, state, and local duties/ taxes on exported products. Advance Authorization Scheme ("AAS") and the

Export Promotion Capital Goods ("EPCG") Scheme are being implemented to enable duty free import of raw materials and capital goods for export production.

GST Regime:

With the introduction of GST regime, there have been many structural changes to the business scenario such as elimination of cascading tax effect, fewer compliances, simple and online procedure and regulation of unbranded players.

3.6.4. Private Sector Investment

Private sector investment, especially Foreign Direct Investment (FDI) and Private Equity (PE), plays a key role in driving growth in India's wire and cable industry, particularly through the real estate sector. Construction consistently ranks among the top three industries attracting FDI. From April 2000 to March 2021, cumulative inflows reached around USD 50.8 billion. In 2024, equity investments in Indian real estate reached a record USD 11.4 billion, reflecting a 54% increase year-on-year. While foreign capital from countries like Singapore, the US, and Canada remains strong, domestic investors led the way, contributing 70% of the total, with developers making up 44% of this investment.

This surge in investment is closely linked to increasing urbanization and a rise in high-end residential and commercial projects. As urban incomes grow and mortgage terms improve, the demand for dependable, high-quality electrical infrastructure increases. In 2023, about 600,000 urban

3.6. 4.. New Age Sectors Driving Demand

Renewable Energy

India's push toward renewable energy is not just about scale. It's changing the demand for more specialized cables. The country aims to install 500 GW of renewable capacity by 2030, which requires over $30 billion in investments. As of January 2025, installed renewable capacity was at 217.62 GW. This means over 310 GW still needs to be added in just six years

. This massive growth is driving a strong need for high-performance solar and wind cables, such as E-Beam cross-linked cables and HV/EHV transmission lines. The PM Surya Ghar rooftop solar scheme has seen over 1 million installations, significantly increasing the need for UV-resistant DC strings. To put this in perspective, each 1 MW solar project uses about 50 km of solar cable. So, the 100 GW solar target alone translates to more than 5 million km of cable.

The power utility sector currently uses around 30% of all wires and cables. In 2023 alone, copper power cable usage reached 70.8 kt, while aluminium hit 404.5 kt. This changing demand is pushing Indian cable manufacturers to invest more in research and development, focusing on durability, safety, and energy efficiency. This is paving the way for a smarter, more future-ready cable industry.

Electric Vehicles (EVs) and Charging Infrastructure

The rapid growth of India's Electric Vehicle (EV) ecosystem is creating a new market for the wire and cable industry. The government aims to install 2 million EV charging stations by 2030, which will drive up the demand for heavy-duty, high-current charging cables. These stations will require around ₹16,000 crore in investment, highlighting the extensive infrastructure development ahead.

EVs also change the demand for cables significantly. Unlike traditional vehicles, electric vehicles need nearly three times more wiring, particularly for battery management, sensors, and power delivery. Consequently, the in-vehicle wiring harness market is expected to grow from USD 4.93 billion in 2024 to USD 6.25 billion by 2035. This change is prompting cable manufacturers to innovate and diversify by creating specialized cables that can support fast charging, resist heat, and fit into tight spaces. The rise of EVs is not just increasing volume; it is also raising standards for performance, safety, and efficiency. This creates long-term opportunities for those in the wire and cable sector who are ready to adapt to the electric revolution.

Data Centres

Data centres are quickly becoming one of the largest growth drivers for the wire and cable industry because of their huge energy and connectivity needs. As India's digital economy grows, data centre capacity is expected to rise from 1.4 GW to 9 GW by 2030. This will lead to a sharp increase in power consumption, growing from under 1% today to about 3% of the country's total electricity usage by the end of the decade.

This growth is causing a spike in demand for high-performance cabling. Worldwide, cabling needs in data centres are projected to nearly triple, increasing from 264.8 kt in 2024 to 727 kt by 2029. India is becoming a significant player in this expansion due to fast infrastructure development and a booming tech ecosystem.

In data centres, wires and cables usually make up 8% to 10% of total capital spending. Their requirement for fault-tolerant, high-redundancy (2N/2N+1) systems mean only top-quality, reliable cabling will suffice. This is driving manufacturers to use better materials, precision engineering, and strict quality standards. Consequently, data centres are not just increasing demand; they are raising the overall expectations for cable performance and reliability in India.

5G Rollout

India's 5G rollout is driving an unprecedented demand for optical fibre cables. Fib erization is essential for the country's digital future. Since October 2022, over 469,000 5G base stations have been installed, but only about 35% of towers are currently fiberized. Telecom operators aim for nearly 100% fibre backhaul. They plan to deliver 1 Gbps connectivity to 120 million households, creating immense growth opportunities. This infrastructure push is leading to a sustained, multi-year demand for optical fibre cables, especially armoured and ribbon types suitable for high-capacity networks. In 2024 alone, India added 1.17 million new Fiber-to-the-Home (FTTH) connections, which shows the growing momentum.

The low fiberization rate reveals a large untapped market for OFC manufacturers. Additionally, 5G's unique needs, such as low latency and high speed, are increasing the demand for specialized fibre types, like bend-insensitive fibres that are perfect for dense urban areas. As India's digital backbone grows, fibre-optic cables will play a central role in this change, providing long-term opportunities for companies in the wire and cable industry.

3.6.5. Continued growth of residential real estate sector

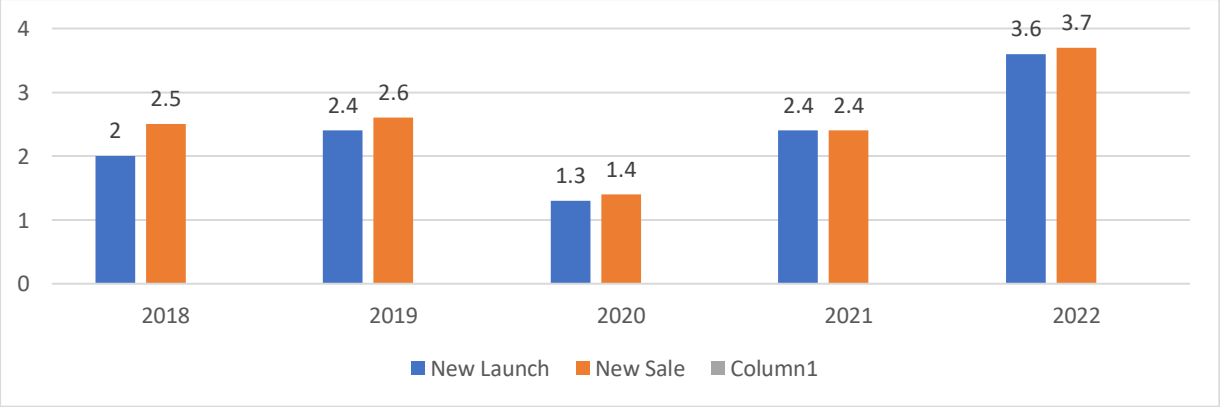
Every newly built residential unit implies not only increased demand for W&C and FMEG products like light, fans and switches, but also creates an installed base for replacement demand for these products. The demand for residential real estate in India depends on various factors such as population growth, economic conditions, and lifestyle changes. With the rise in population and urbanization, the demand for houses is likely to increase. India is expected to witness the growth in residential real estate market driven by both, the private sector led residential real estate and of government led aimed at affordable housing (PMAY and others).

India's residential real estate is growing rapidly due to affordable financing, a demand for premium housing, and strong launches, particularly in developing metro and Tier-2 cities. This growth boosts the wire and cable sector by increasing volumes per dwelling, raising wiring standards in luxury homes, and increasing demand for smart-home and fibre infrastructure. As major companies increase production, the residential sector stays crucial in pushing the wire and cable industry forward.

With the increase in demand for the residential space in both affordable and premium segments, India's residential market has witnessed a rapid growth in recent years. Residential sales volume across the major cities in India witnessed a 51% annual rise from approximately 2.4 lakhs units sold in CY 2021 to approximately 3.7 lakhs unit sold in CY 2022 signifying a healthy recovery in the sector, post COVID-19 pandemic. The top 7 Indian cities with the highest residential unit sales in CY 2022 are Mumbai, National Capital Region (NCR), Bengaluru, Pune, Chennai, Hyderabad, and Kolkata.

Supply of Residential House in top 7 cities of India (in Lakhs) (CY)

Table 5:Supply of Residential House in top 7 cities of India (in Lakhs) (CY)



Source: RR Kabel DRHP

The GoI has implemented various reforms in the real estate sector to improve transparency, affordability, and accessibility of housing such as RERA and affordable housing policies which has significantly increased the confidence of the home buyers and investors. Further, the introduction of GST has simplified the tax structure for the real estate sector, reducing the overall tax burden on home buyers. There are many more such housing board schemes being run by Indian state governments. These combined efforts of both the central and state governments to facilitate a greater number of residential houses is expected to continuously create and increase demand for W&C and FMEG products in the coming years.

Resilient commercial real estate sector

India's commercial real estate sector is booming, driven by strong economic growth, new industries, changing workplace norms, and policies that support investors. This rise in commercial real estate is a significant growth driver for the wire and cable industry. As smart, certified cables are increasingly used in Grade-A buildings, data centres, and logistics hubs, the demand for traditional power cables and fibre infrastructure is growing rapidly. The wire and cable sector are well-positioned to take advantage of this commercial real estate trend, increasing production and innovation to meet the rising needs for commercial infrastructure. The commercial property market is estimated to expand at a CAGR of almost 13% from the projected period of the CY 2022 to CY 2027. The commercial real estate sector mainly comprises of office space, retail space, data centre space in both owned and leased premises. India's commercial real estate (CRE) sector is experiencing unprecedented growth. It is set to nearly quintuple in value, increasing from around USD 50 billion in 2024 to over USD 250 billion by 2033, showing a strong 20% CAGR. This solidifies India's place as the most resilient CRE market in Asia-Pacific.

This remarkable growth is fuelled by several key factors:

Robust Economic Growth: India's booming economy is on track to become the world's third largest by 2029. This growth drives demand across all CRE segments.

Demographic Dividend and Urbanization: A young, urban population with rising disposable incomes consistently creates demand for modern commercial spaces.

Thriving Business Ecosystems: A dynamic startup scene, along with growth in IT/Its, BFSI, and Global Capability Centres (GCCs), maintains a steady demand for modern office spaces.

Increased Foreign Investment: Strong foreign direct investment (FDI) inflows reflect international confidence in India's real estate potential.

Supportive Government Policies: Reforms such as RERA and GST, along with major infrastructure projects, improve transparency and attractiveness.

Flexible Work Models: The growing popularity of co-working and managed offices supports the rise of hybrid work.

Sustainability Focus: There is an increasing demand for green-certified, smart buildings that include IoT and automation.

E-commerce and Manufacturing Boom: This surge drives the need for modern retail spaces and advanced warehousing facilities.

Growth Hotspots: Major cities like Bengaluru, Hyderabad, Delhi-NCR, Mumbai, and Chennai lead the way. Additionally, Tier II and III cities are quickly becoming important growth centres, especially for logistics.

Impact on the Wire and Cable Industry:

The CRE boom is greatly benefiting the wire and cable sector: Surging Demand: Each new commercial development requires vast amounts of wires and cables for power, lighting, and communication networks.

Demand for Specialized Cables: Safety: There is an increased need for flame-retardant, low-smoke (FR-LSH/LSZH) cables. Connectivity: The demand for advanced communication cables, particularly optical fibre, is high. These cables are essential for smart buildings and data centres and are projected to exceed USD 5 billion by 2030.Efficiency: There is a growing need for energy-efficient cables.

3.6.6. Transition of automobiles and transport towards EVs

The fast shift to electric vehicles in India will change the game significantly, affecting not only the automotive industry but also related sectors like the Wire & Cable (W&C) industry. As electric vehicles (EVs) become more common, the need for strong charging infrastructure, both public and private, is increasing sharply. This rise is directly tied to a higher demand for electrical products.

The Indian government has set a bold goal to achieve 30% electrification of the vehicle fleet by the calendar year 2030. This effort is supported by significant policy backing and incentives, acknowledging that EVs are vital for cutting fuel use and promoting sustainable transport. EV sales are expected to increase four times by the calendar year 2025 and surpass 10 million units annually by the calendar year 2030. Overall, India's domestic EV market is projected to grow at an impressive 47% annual rate between the calendar years 2022 and 2030, expanding from US$ 3.2 billion to about US$ 110 billion by the calendar year 2029.

3.7. Key Government Initiatives Driving EV Growth

3.7.1. FAME India Scheme (Faster Adoption and Manufacturing of (Hybrid and) EVs):

Launched in 2015 and extended until CY 2024, FAME-II has a budget of US$ 1.3 billion. This program offers incentives for e-two-wheelers, three-wheelers, passenger vehicles, and buses. It aims to encourage early adoption and manufacturing.

3.7.2 Production Linked Incentive (PLI) Scheme for Advanced Chemistry Cell (ACC) Battery Storage:

This plan seeks to increase local battery manufacturing. India's battery storage capacity is expected to hit 600 gigawatt hours (GWh) by calendar year 2030.

3.7.4 Battery Swapping Policy:

Introduced in the 2022-23 budget, this policy aims to create an efficient battery swapping system with interoperability standards. This will lower upfront costs for electric vehicles and encourage adoption, particularly for commercial vehicles.

3.7.5. Other Incentives:

These include tax exemptions on EV loans, reduced customs duty on nickel ore, and a cut in GST on electric car sales from 12% to 5%.

Beyond government efforts, the e-commerce and logistics sectors are actively contributing. Companies like Flipkart and Zomato have promised to fully switch to electric vehicles (EVs) for their delivery fleets by calendar year 2030. This move will further increase demand.

This widespread adoption of EVs creates a huge need for EV infrastructure. It includes mobility solutions, energy, and essential electrical products like wires and cables. India is expected to need the installation of 2.05 million charging stations by calendar year 2030. This rise in demand for charging stations, whether at public points, workplaces, or homes, requires a strong supply of specialized wiring, cables, and switches. This makes the EV revolution a lasting driver of demand for the wiring and cabling industry.

3.7. 6.. Rural Electrification

The better road connectivity in India's villages and rural areas has greatly raised the standard of living for rural households. This improved connectivity has increased the demand for essential electricity services, which has boosted the need for new electrical goods.

The Indian government has been proactive in addressing rural electrification through key initiatives:

Deendayal Upadhyaya Gram Jyoti Yojana (DDUGJY):

Launched in December 2014, DDUGJY aimed to supply electricity to rural India and to electrify villages that previously had no power.

Pradhan Mantri Sahaj Bijli Har Ghar Yojana – Saubhagya:

Introduced in October 2017, Saubhagya aimed to provide electricity connections to all rural and urban poor households that wanted them. The goal was universal household electrification.

These programs have produced significant results. According to the Ministry of Power, a total of 2.9 crore households were electrified by March 2023. Notably, seven states—Assam, Chhattisgarh, Jharkhand, Karnataka, Manipur, Rajasthan, and Uttar Pradesh—reported 100% household electrification as of March 31, 2021. While major progress has been made, around

1.18 million households still need to be electrified, with completion expected in the next few years.

The journey of rural electrification in India has been impressive. The number of electrified villages grew from just 3,000 in 1950 to 600,000 villages by 2022. This widespread access to electricity is a key driver of demand for the wire and cable industry. Every new connection and household appliance requires dependable electrical infrastructure. Ongoing efforts to electrify the remaining households and improve supply quality in rural areas will continue to create steady demand for wires, cables, and other electrical products. This will greatly support the growth of the wire and cable industry.

Electrification in India has increased over time with the support from schemes like DDUGJY, Saubhagya and IPDS, which has given the electrical market a major boost. It has opened new markets for companies in the W&C and FMEG industry, that can now sell their products into the rural parts of India as well. The expansion of rural electrification is expected to significantly increase the demand of electrical products, as new power lines and infrastructure need to be set up and maintained.

Number of villages electrified in India (CY)

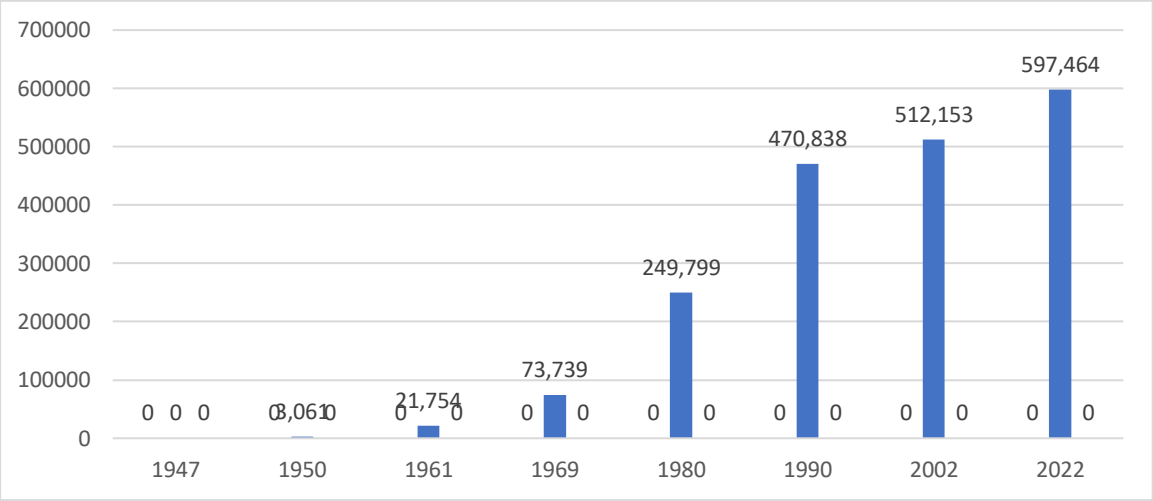


Figure 13:: Number of villages electrified in India (CY)

Source: PIB

3.7.7 Push towards Renewable Energy

In line with the Paris Agreement's goal to keep temperature rises well below 2°C, countries around the world are working towards cleaner and more sustainable environments. Japan, South Korea, and the European Union have set goals to reach net-zero emissions by CY 2050. India aims to achieve net-zero emissions by CY 2070. These commitments are speeding up the global move toward renewable energy, like solar and wind power. This shift is becoming an important growth driver for the Wire & Cable (W&C) and Fast-Moving Electrical Goods (FMEG) industries, both in India and around the world.

India's energy demand is set to grow significantly in the coming decades. This makes the push for renewable energy crucial. The country aims to install 500 GW of renewable energy capacity by CY 2030. This is expected to require an investment of at least ₹2.4 lakh crore, or about US$ 28.8 billion. By February 2023, India's renewable energy capacity had already reached 168.96 GW. A large part of the 2030 target, 280 GW, is expected to come from solar energy.

Solar power is growing rapidly. As of November 30, 2022, India had about 62 GW of installed solar power capacity. The country is working towards the 100 GW target set by the National Solar Mission (NSM), which started in January 2010. This capacity includes 52 GW from ground-mounted projects, 7.8 GW from rooftop solar, and 2.1 GW from off-grid solar. The ongoing approval of new solar projects in various states is increasing the demand for electrical goods, particularly for connecting long-distance grids.

To support these large-scale solar projects connected to the grid, the "Development of Solar Parks and Ultra Mega Solar Power Projects" scheme is being actively carried out. It aims for a target capacity of 40 GW by March 2024. This major increase in renewable energy infrastructure requires a lot of cabling for power transmission, distribution, and connections within solar parks and to the main grid. This directly boosts demand for the W&C industry. The move towards green energy is not only an environmental necessity but also a major economic opportunity, greatly affecting the growth of electrical product manufacturers.

Solar energy capacity forecast (Fiscal year)

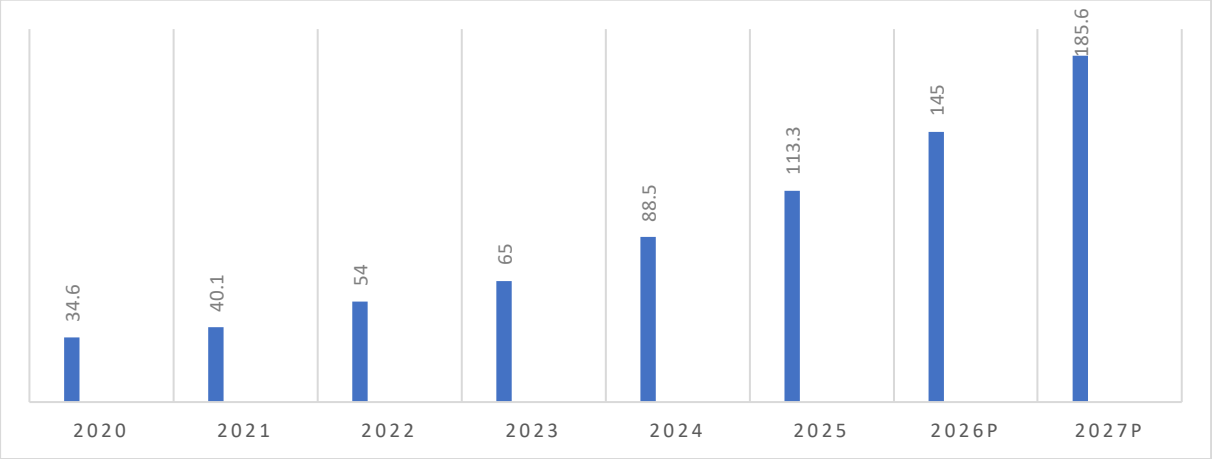


Figure 14:Solar energy capacity forecast (Fiscal year)

Source: RR Kabel DRHP

Table 6:the impact of the key growth drives on W&C and FMEG sector.

Impact of key growth drivers on the W&C and FMEG industry

The table below depicts the impact of the key growth drives on W&C and FMEG sector.

Sl. No.	Key Growth Drivers of W&C and FMEG industry	W&C	Lights	Fans	Switches
1	Public and private investment outlay in infrastructure	High	Medium	Medium	Medium
2	Urbanization and smart cities	High	Medium	Medium	Medium
3	FDI and PE investment in real estate sector	High	High	**High**	High
4	Continued growth of residential real estate sector	High	High	High	High
5	Resilient commercial real estate sector	High	High	High	High
6	Transition of automobiles and transport towards electric vehicles	High	NA	NA	NA
7	Policy reforms promoting production, exports and ease of doing business	High	Medium	Medium	Medium
8	Digitalization	High	NA	NA	NA
9	Safety standards	High	High	High	High
10	Rural electrification	High	Medium	Medium	Medium
11	Push towards renewable energy	High	NA	NA	NA

Chapter 4 : Wires and Cables Market in India

4.1. Industry Overview

The wires and cables (W&C) sector is important to India's consumer electrical industry. It made up 41% of the overall market in FY23. The industry has grown from ₹33,500 crore in FY15 to ₹74,800 crore in FY23, showing a healthy CAGR of 11%. This growth is supported by the expansion of infrastructure and housing in India.

In FY24, the W&C market is expected to be worth about ₹77,600 crore (USD 9.32 billion). While different estimates may vary, this number is close to historical growth and the Technopak baseline. By FY27, the market is forecasted to reach ₹1,20,000 crore, with growth speeding up to a 13% CAGR, indicating a faster growth phase.

This momentum comes from government initiatives such as the National Infrastructure Pipeline (NIP), PMAY, Smart Cities, and renewable energy targets. As these programs move into execution, they create strong and steady demand for essential components like W&C, setting the industry up for ongoing growth.

Indian Wires and Cables Market – By Value (₹ Cr) in Fiscal Year

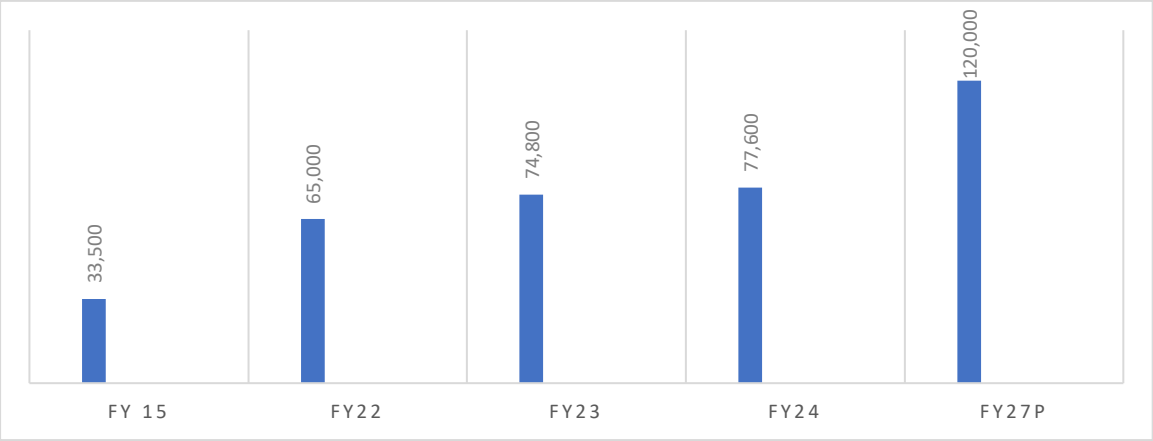


Figure 15:Indian Wires and Cables Market – By Value (₹ Cr) in Fiscal Year

Source: RR Kabel DRHP, Samar wealth

As a result of infrastructure development, urbanization, smart cities, the growth of residential and commercial real estate, rural electrification, and the move towards renewable energy, the cables & wires segment is expected to experience significant growth. The cables & wires market reported ~8% CAGR over FY14-23 to reach INR700b and it is estimated to clock 12-14% CAGR over FY23-27 (INR1.2-1.3t). The industry growth rate was hit by the challenges associated with Covid-19 in FY20-21 and it recorded a CAGR of mere 2.7% during FY19-22.

4.2. Share of W&C within Consumer Electricals

In FY23, wires and cables (W&C) made up 41% of India's ₹1.81 lakh crore consumer electrical industry. Fast-moving electrical goods (FMEG) accounted for the remaining 59%.

Looking ahead, W&C is expected to grow at a faster pace, with a 13% CAGR compared to 8% for FMEG through FY27. This indicates a key phase in India's development, where growth

driven by infrastructure is taking the spotlight. The rising demand for W&C shows ongoing investments in power grids, housing, and industrial expansion. These projects require cabling before consumer products are introduced.

This infrastructure development not only fuels W&C growth today but also sets the stage for future FMEG demand. New homes, factories, and commercial spaces will eventually need fans, lighting, and appliances. In short, the increase in W&C is a leading sign of India's next wave of consumption.

Indian Wires and Cables Market – By Value (₹ Cr) in Fiscal Year

Table 7:Indian Wires and Cables Market – By Value (₹ Cr) in Fiscal Year

Segment	FY23 Value (₹ Crore)	FY27(P) Value (₹ Crore)	CAGR (FY23-FY27P)
Wires and Cables (W&C)	74,800	1,20,000	13%
Fast Moving Electrical Goods (FMEG)	1,06,350	1,46,500	8%
Total Consumer Electricals	1,81,150	2,66,500	10%

Source: RR Kabel DRHP, samar wealth

India's per capita electricity consumption was just 1,255 kWh in 2021, far below the global average of 3,700 kWh. This gap shows a significant untapped potential as more households and businesses gain access to power.

As urbanization increases and incomes rise, electricity usage is expected to grow, increasing demand for wires and cables (W&C). From basic housing wires in new homes to high-capacity cables for improved power infrastructure, the sector is set to benefit at every level.

In short, low current consumption isn't a weakness; it's an opportunity. As India increases its power supply, the W&C industry will be one of the biggest winners.

4.3. Market Segmentation by Type

The W&C market is divided into five main sub-categories. Each one has different applications and plays a unique role in the industry: Housing Wires, Power Cables, Control and Instrumentation Cables, Communication Cables, and Flexible and Specialty Cables.

4.3.1. Housing Wires

Housing wires are the largest and fastest-growing part of India's wires and cables (W&C) market. They accounted for nearly ₹24,535 crore or 32.8% of the market in

FY23. This share is expected to increase to 34.6% by FY27, supported by a robust 14.1% CAGR.

Their common use in homes, offices, and commercial buildings makes them a clear sign of India's construction and urban growth. In simple terms, as India builds more, the demand for housing wires rises.

Government programs like PMAY, smart city initiatives, and the booming commercial real estate sector, which includes malls and data centers, are driving this growth. RR Kabel leads the segment with a 10.4% market share, followed by Finolex. The data clearly shows that the foundation of India's W&C market is being built, wire by wire.

4.3.2. Power Cables (LT/HT/EHV)

Power cables, including low, high, and extra-high voltage lines, accounted for 24.6% of India's wires and cables market in FY23. This share is expected to remain stable through FY27 with a 12.5% compound annual growth rate.

These cables are essential for India's power transmission and distribution systems. Their demand is growing due to two main trends: the modernization of the national grid and the rise of renewable energy. Their use in solar and wind projects connects directly to India's green energy goals.

What stands out is this: while housing wires reflect growth for end users, power cables indicate the country's effort to strengthen its energy infrastructure. India isn't just using more electricity; it's improving how that electricity is delivered. Polycab and KEI are the key players likely to benefit from this ongoing change.

4.3.3. Control and Instrumentation Cables

Control and instrumentation cables made up 3.4% of the wire and cable market in FY23. They are expected to rise to 3.5% by FY27, growing at a steady rate of 8.7%.

Though small in size, this segment plays a key role in industrial automation. These low-energy signal cables are vital for monitoring and controlling power systems in industries like chemicals, automotive, steel, and oil and gas.

Their moderate growth shows the steady progress of industrial modernization instead of large infrastructure projects. Unlike housing or power cables, this segment grows quietly in the background. It supports the precision, safety, and efficiency that modern manufacturing requires.

In short, control cables may not make headlines, but they are essential for industries that prioritize consistency, automation, and reliability.

4.3.4. Communication Cables

Communication cables made up 14.6% of India's W&C market in FY23. They are expected to grow at a strong rate of 13.4% annually through FY27, even as their share slightly drops to 12.7% because other segments are growing faster.

These cables, which include fiber optic, ethernet, and coaxial types, are essential to India's digital infrastructure boom. The major drivers are the 5G rollout and the BharatNet initiative, which aims to provide high-speed internet to rural areas. Although there are over 4.6 lakh 5G towers, only 35% have fiber connections, indicating a significant opportunity for fiber deployment.

India aims for 1 Gbps connectivity for 120 million homes and leads in fiber production, with over 100 million fiber-km expected in FY24. This shows that the country is not only consuming high-tech communication cables but also exporting them. The growth in this segment reflects a broader change, shifting from building infrastructure to supporting a digitally connected economy.

4.3.5. Flexible and Specialty Cables

Flexible and specialty cables accounted for 24.6% of India's wires and cables market in FY23, and this share is expected to remain stable through FY27, driven by a strong 12.5% CAGR.

These versatile cables are used in many industries, including consumer appliances, mining, railways, marine, and oil and gas. Future growth is fueled by new areas like electric vehicles (EVs) and Multimodal Logistics Parks under the PM Gati Shakti plan. For instance, EVs need nearly three times more wiring than traditional vehicles, creating a significant new growth opportunity.

The rise of this segment indicates a maturing economy with increasingly diverse electrical needs. As India upgrades its transport, logistics, and industrial base, the demand for specialized, high-performance cables is growing. The diversification across sectors also makes this segment more resilient, reducing its reliance on housing or power cables alone and establishing it as a key part of India's changing electrical landscape.

Share of different products within the cables & wires market (FY23)

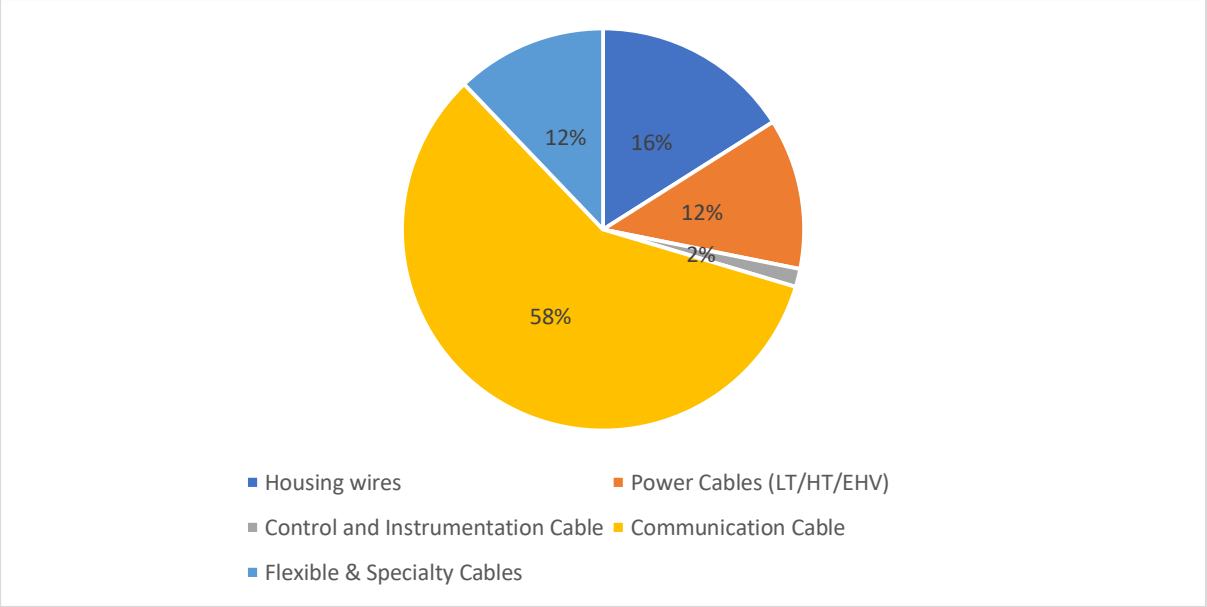


Figure 16:Share of different products within the cables & wires market (FY23)

Source: MOFSL

The domestic cables & wires market has steadily moved to an organized play from a largely unorganized play, including regional/national players. The shift was led by rising awareness among consumers towards safety and quality, the advent of GST regime, technological innovation, brand building, and technical & regulatory compliance.

Sub-categories of Wires and Cables with examples

Table 8:Sub-categories of Wires and Cables with examples

Sub-category of Wires & Cables	Sectors in which pre-dominantly used	Examples
Housing Wires	• Wiring of residential and commercial buildings such as apartments, shopping complexes and offices.	
Power Cables	• Power transmission and distribution • Solar and wind energy	
Control and instrumentation cables	• Chemical and fertilizers • Automotive • Steel • Oil and gas	
Communication cables	• Service sector • Industrial sector such as steel, automotive etc. • Residential buildings for internet, intercom purpose	
Flexible and specialty cables	• Consumer appliances • Railways • Mining • Marine, oil and gas facilities offshore / onshore	

Source: RR Kabel

Table 9:Sub-category wise CAGR of W&C between Fiscal 2023 and Fiscal 2027(P)

Sub-category of Wires & Cables	CAGR between Fiscal 2023 and Fiscal 2027(P)
Housing Wires	14.1%
Power Cables	12.5%
Control and instrumentation cables	8.7%
Communication cables	13.4%
Flexible and specialty cables	12.5%

Source: RR Kabel

4.4. Trends shaping the Domestic Wires and Cables Market

4.4.1. Shift towards Branded Play

As of Fiscal 2023, branded players have nearly 72% of the market share (approximately ₹ 53,860 crore) of the W&C market in India. Within this 72%, five leading players namely Polycab, KEI, Havells, Finolex and RR Kabel, garner approximately 60%-62% market share and the balance 38%-40% is constituted by challenger brands like Syska and V-Guard. Branded players are estimated to capture approximately 80% (approximately ₹ 96,000 crore) market share by Fiscal 2027.

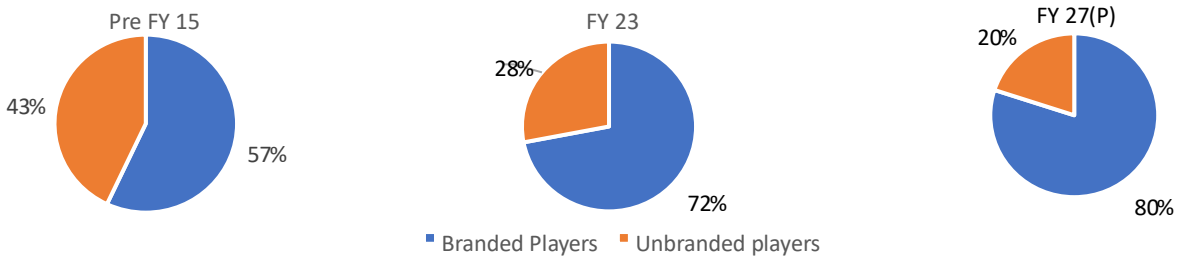


Figure 17:Share of Branded Players in Indian W&C Market – By Value (Fiscal Year)

The W&C market in India has steadily moved from largely unbranded play towards branded play including regional and national players, because of reasons such as rising awareness among consumers towards safety and quality, advent of GST regime etc. Apart from these, the consolidation of branded play of the wires and cables market over unbranded play is driven by two other facts.

4.4.2. Rising Consumer Awareness and Quality Imperatives

As consumers become more informed and safety-conscious, there is a noticeable shift away from cheap, unbranded electrical products toward trusted, branded alternatives. People are no longer willing to compromise on quality, especially regarding risks like fire hazards, toxic materials, or electrical failures. This growing awareness is driving demand for branded wires and cables, which people see as more reliable and safer in the long run. Recognizing this trend, leading brands are not just meeting expectations; they are actively educating consumers about the dangers of substandard products. Through well-crafted marketing campaigns, they highlight the importance of features like fire resistance and energy efficiency, effectively positioning their products as worth the extra cost. For many buyers, that added expense is justified not just by better materials but by peace of mind and long-term reliability. As a result, safety and durability are becoming central to purchasing decisions, reinforcing the value and market strength of branded options.

4.4.2. Impact of the GST Regime

The rollout of the Goods and Services Tax (GST) has changed the Indian economy, especially for the wires and cables (W&C) sector. By bringing more transactions into the formal economy, GST has narrowed the gap between branded and unbranded players. Many unbranded manufacturers used to work in the informal sector, avoiding taxes and regulations. This allowed them to keep prices low. However, with GST promoting more transparency and accountability in the supply chain, these manufacturers now face higher costs for compliance, which reduces their pricing edge. On the flip side, branded manufacturers have gained from a simpler and more efficient tax system. This helps them optimize their operations and lower costs. As a result, the price difference between branded and unbranded products has shrunk, making it tougher for informal manufacturers to compete. This change has not only boosted the competitiveness of organized brands but is also leading to a gradual consolidation of the market, where compliant, quality-driven players are gaining more ground.

4.4.3. Strategic Brand Building through ATL Marketing

Branded players in the wires and cables (W&C) industry are investing heavily in Above-The-Line (ATL) marketing to create strong brand recognition, gain consumer trust, and stand out in a crowded market. These campaigns often focus on key themes like safety, durability, and innovation. They use mass media channels such as television, print, outdoor advertising, and increasingly, digital platforms. Celebrity endorsements boost brand recall and credibility. Polycab India's campaign featuring youth icon Ayushmann Khurrana is a prime example, using music to connect emotionally with consumers. However, the strategy extends beyond just ATL. Leading brands like Polycab also engage actively on social media, combining traditional advertising with digital interaction and localized Below-The-Line (BTL) initiatives. This integrated, Through-The-Line (TTL) approach resembles what global brands like Amazon and Cadbury do. It creates a strong and consistent brand presence across different platforms. This strategy is complex and resource-intensive, making it difficult for smaller, unbranded players to compete. Ultimately, this complete marketing mix doesn't just raise awareness; it builds lasting consumer connections and strengthens the premium positioning of branded products in the W&C market.

4.4.4. Robust Distribution Networks and Market Reach

Branded companies in the wires and cables industry have built strong, wide-ranging distribution networks that cover the entire country. This includes major cities, smaller towns, tier III cities, and rural areas. Such extensive infrastructure gives them a clear advantage, allowing for reliable product availability and customer service across different regions. However, this broad reach is not just about being present; it requires considerable scale, complexity, and investment to create and sustain. This involves setting up warehouses, managing transportation fleets, training a large sales force, and forming strong partnerships with retailers nationwide.

In contrast, unbranded companies often face limitations due to regional operations and smaller budgets. They simply cannot match this level of market presence. As a result, branded firms are in a better position to attract new customers and drive growth, while the lack of similar infrastructure continues to hinder the expansion of unbranded competitors. This solid and reliable distribution network not only boosts the market position of branded players but also creates a significant barrier to entry, speeding up the shift towards organized, branded leadership in the industry.

4.5. Technological Innovations and Product Differentiation

Branded players in the wires and cables industry are constantly pushing boundaries with ongoing product innovation and the integration of new technologies. These efforts have led to cables that not only perform better but also provide important benefits like increased safety and improved energy efficiency. Such innovations create clear and meaningful differences, setting branded products apart in a market where quality and reliability are crucial. For unbranded players, keeping up with this pace of change is a significant challenge since they often lack the resources or expertise to match these advancements. As a result, technological progress has become a strong competitive barrier that reinforces the value of branded products. This makes it tougher for low-cost, unbranded options to compete.

4.5.1. LS0H (Low Smoke Zero Halogen*)*

Wires and cables with LS0H (Low Smoke Zero Halogen) insulation are designed with safety in mind. These products use special jacketing materials that produce little smoke and no toxic halogens when exposed to high heat or fire. This is especially important in enclosed or poorly ventilated areas like aircraft, trains, or underground facilities. Thick smoke and harmful gases can greatly reduce visibility and create serious health risks during emergencies. Understanding the need for this technology, all five leading companies in India's wires and cables industry—Polycab, KEI, Havells, Finolex, and RR Kabel—now provide LS0H products. Notably, RR Kabel was the first company in India to offer this insulation, setting a new standard for fire safety in the industry.

4.5.2. UCT (Unilay Core Technology)

Wires and cables made with Unilay Core Technology (UCT) provide several practical benefits beyond just electrical performance. Their high-density wiring helps save valuable space, which is important in compact installations. UCT also makes products more durable during handling. It reduces the chances of breakage when stripping

insulation and ensures cleaner, more secure connections through compact bunching. This improves conductivity and results in material savings. Additionally, the increased mechanical strength of UCT products reduces the risk of individual strand breakage, leading to greater safety and reliability. RR Kabel was the first company in India to use UCT in its heat-resistant and flame-retardant wires and cables. This move continues to set the pace for innovation in the industry.

4.5.3. E-beam (Electron Beam Cross-linking)

Electron beam cross-linking technology is a new development in wire and cable manufacturing. It offers excellent insulation protection, especially in extreme situations like short-circuits. This process improves several important performance traits, such as tensile strength at high temperatures, resistance to abrasion and heat, as well as cut-through, shear, and compressive strength. As a result, cables treated with this technology are more durable, reliable, and safe in tough environments. Furthermore, electron beam cross-linked wires and cables are seen as environmentally friendly, meeting the rising demand for sustainable solutions. Among the industry leaders, Polycab stands out for producing e-beam cables. Other notable players, like Apar Industries and Vindhya Telelinks, are also adopting this eco-friendly technology.

These technological advancements in the wires and cables industry are not just minor upgrades. They represent significant investments in research, new materials, and modern manufacturing. Leading brands like RR Kabel and Polycab have been at the forefront. RR Kabel is known for its LS0H and Unilay Core Technology, while Polycab has made strides in electron beam (e-beam) innovation. Their early use of these technologies shows a strong commitment to staying ahead through innovation. This dedication creates a powerful cycle. As these companies develop better, unique products, they can charge higher prices and grow their market share. In turn, the increased revenues help fund more research and development, attract top talent, and invest in the latest machinery. This continuously widens the gap between them and their unbranded competitors. Unbranded players, often limited by money and expertise, find it hard to keep up. This makes it increasingly tough to compete on technology or performance. This ongoing cycle strengthens the long-term advantage of branded players and reinforces their leadership in the industry.

4.6. Increase in per capita per annum electricity consumption

The global average of per capita per annum electricity consumption is estimated to be approximately 3700 kWh in CY 2021. India's per capita per annum electricity consumption is low, as compared to other countries like USA and China, at approximately 1255 kWh, there by suggesting headroom for growth in future. Such increase in per capita per annum electricity consumption is expected to drive the demand for W&C products like housing wires in India.

4.7. Key Players in the Industry

The Indian W&C industry has been gradually moving from largely unbranded play towards branded play including regional and national players, as a result of rising awareness among consumers towards safety and quality, advent of GST regime, increasing technological and product complexities, as well as growing marketing and branding activities by branded players.

The Indian W&C industry has two main parts: a competitive organized sector and a large, fragmented unorganized sector. The organized sector includes about 15 to 20 large and medium-sized companies. It makes up roughly 65 to 70% of the market by value. The remaining 30 to 35% comes from many small regional players in the unorganized sector.

A strong trend of consolidation is happening, benefiting the organized players. Several factors are driving this change. First, the implementation of the Goods and Services Tax (GST) has improved tax compliance and lessened the pricing edge that unorganized players had when avoiding taxes. Second, the Bureau of Indian Standards (BIS) is enforcing quality standards more strictly. This makes it harder for makers of low-quality products to compete. Third, more consumers are becoming aware of electrical safety. They are increasingly wary of the risks tied to poor-quality wiring, which boosts the demand for branded products, especially in homes. This shift from unorganized to organized is providing steady growth for listed companies. It allows them to take market share and grow faster than the industry as a whole. CRISIL Ratings expects the unorganized sector's share to drop from 34% to 26% in the next five years. This highlights the size of the opportunity.

revenue of Branded Players in Indian W&C Market – By Value (Fiscal Year)

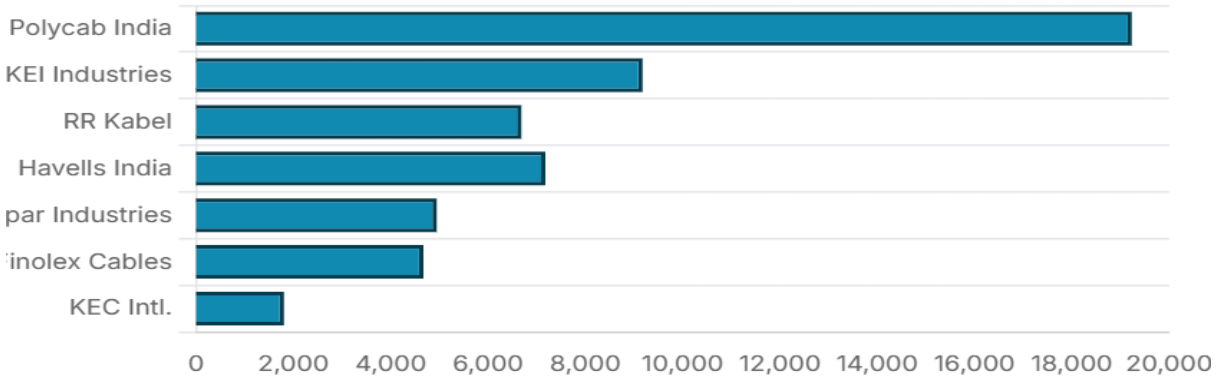


Figure 18:Revenue of Branded Players in Indian W&C Market – By Value (Fiscal Year)

Source: Company Annual Reports (FY24/FY25)

4.8. Profiling the Leaders

The organized market is led by a few major players, each with their own strategic focus and strengths. The arrival of new, well-funded companies like Adani Group and UltraTech Cement shows the sector's appeal but also hints at a possible rise in competition.

4.9. Financial and Economic Analysis

4.9.1. Industry Size and Growth Trajectory

Estimating the size of the Indian Wires and Cables market requires combining data from various sources. These sources often have different methods and scopes, such as whether they include the unorganized sector. A CRISIL Ratings analysis from March 2025 estimates the organized sector's revenue at ₹80,000 to ₹82,000 crore for FY25. If the organized sector makes up about two-thirds of the total market, the overall market size is around ₹1,20,000 to ₹1,24,000 crore (US$14.5 to 15.0 billion) for FY25. Other market research firms offer different estimates.

Mordor Intelligence estimates it at US$21.2 billion, while Fortune Business Insights puts it lower at US$10.0 billion for 2025. This shows the difficulty in accurately measuring the fragmented unorganized segment. For this report, we will use the CRISIL-derived figure as the main benchmark.

The industry has shown strong growth in the past, driven by both volume and value, particularly due to rising commodity prices. Looking forward, a consensus from industry reports and company forecasts suggests continued double-digit growth. CRISIL expects a solid 15 to 16% growth for the organized sector in FY26, fuelled by strong domestic demand and a 20 to 22% increase in exports.

Market Growth by Segment (₹ Crore) in Indian W&C Market – By Value (Fiscal Year)

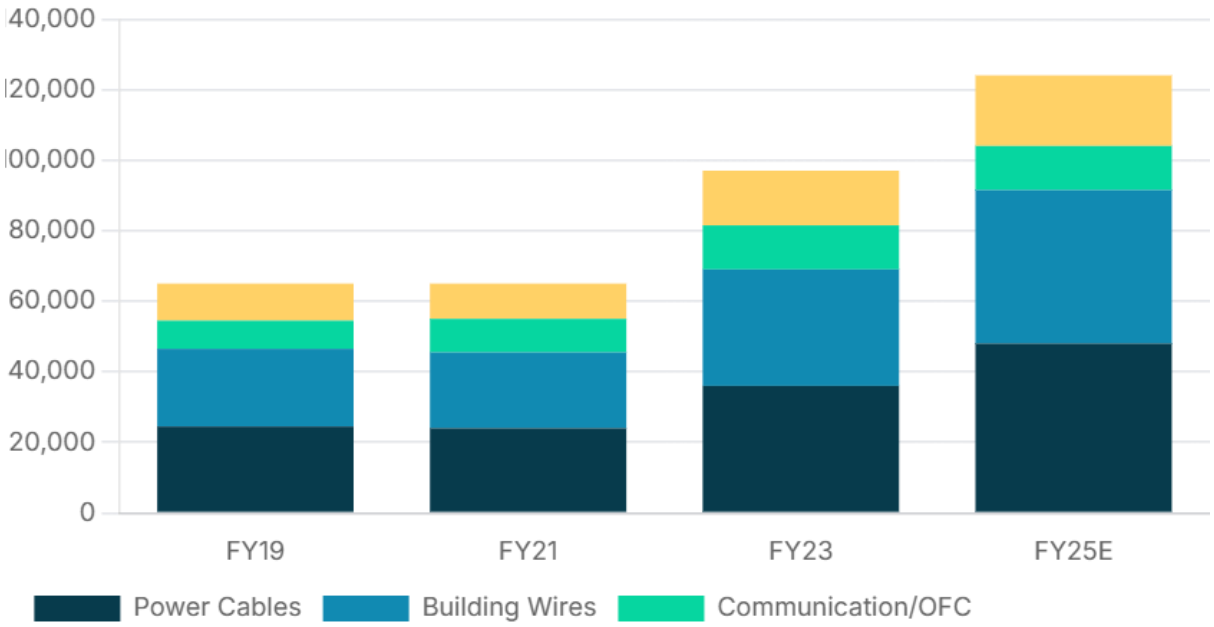


Figure 19:Market Growth by Segment (₹ Crore) in Indian W&C Market – By Value (Fiscal Year)

4.10. Unit Economics and Profitability

The financial health of W&C manufacturers is closely linked to how well they handle changes in commodity prices and manage their working capital.

4.10.1. Raw Material Basket and Margin Impact:

Raw materials, mainly copper and aluminium, can make up 60-70% of a cable's selling price. This high reliance makes industry margins very sensitive to changes in global metal prices on exchanges like the London Metal Exchange (LME). Established companies have solid ways to pass on these price changes to customers, especially in the B2B segment, where pricing often ties to LME rates. However, there is usually a delay of one to two months, which can lead to

short-term margin fluctuations. In the B2C segment, price increases happen more sporadically. The ability to handle this pass-through process well is an important factor for profitability.

4.10.2. LME Copper & Aluminium Prices vs. W&C EBITDA Margins (Quarterly)

Note: A complete graphical representation needs time-series data that is not fully available in the provided snippets. The chart would display LME Copper and Aluminium prices on the primary Y-axis and the quarterly EBITDA margins of Polycab, KEI, and RR Kabel on the secondary Y-axis from Q1 FY19 to Q4 FY25. The expected visual would show margin compression in quarters after sharp commodity price spikes, followed by a recovery as price increases are applied. This demonstrates the pass-through nature of the business.

4.10.3. Working Capital Intensity:

The industry relies heavily on working capital. Manufacturers must keep a large inventory of different raw materials and finished products. In the institutional (B2B/EPC) segment, credit periods can be longer, resulting in high receivables. Managing working capital effectively is essential for success. Companies with a larger B2C mix, such as RR Kabel, which had 56 days in FY25, generally have a more efficient cycle. In contrast, those with considerable EPC involvement, like Universal Cables, had 155 days in FY25. Payments in this sector are milestone-based and can be delayed.

4.11. India's Role in Global Trade

India has successfully become a net exporter of wires and cables. It has improved its manufacturing scale and quality to compete globally. The trade data analysis mainly uses the Harmonized System (HS) code 8544. This code includes most insulated wires and cables, such as OFC found in sub-heading 8544.70.

4.11.1. India's W&C Trade Balance

BandtheimportlineatUS1.29B, showing a clear trade surplus.)

- *Export Dynamics*: In calendar year 2023, which serves as a proxy for FY24, India's exports under HS Code 8544 reached US$2.0 billion. The main destinations are developed economies. This shows that Indian products are well accepted in quality-focused markets. This trend benefits from the global "China+1" supply chain strategy. Multinational companies are looking for alternative manufacturing centres to reduce risks in their supply chains.

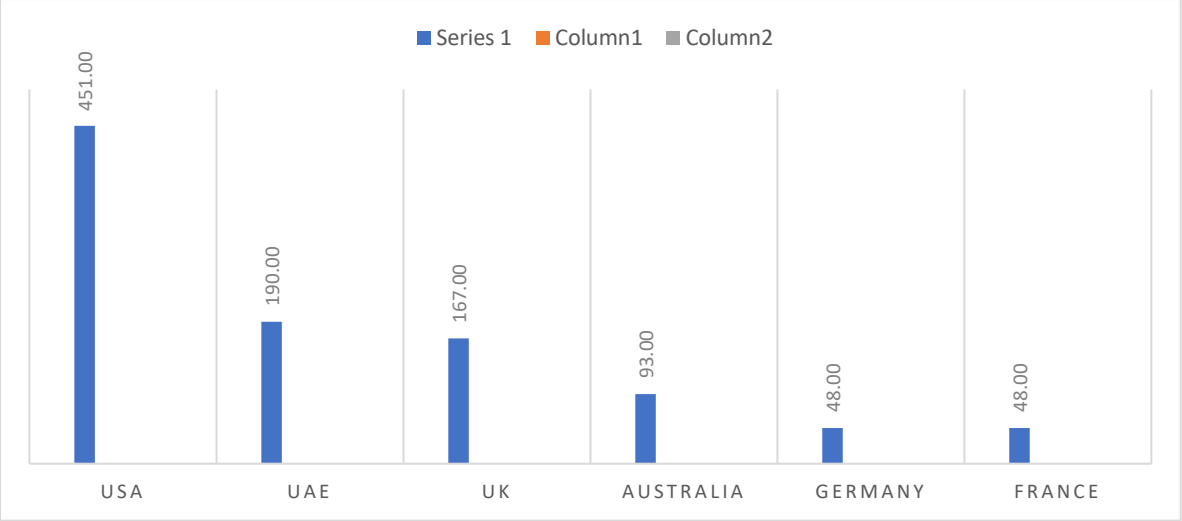


Figure 20:Top 6 Export Destinations for Indian W&C (FY2024)

Source: Trend Economy, data for calendar year

- *Import Dynamics:* India's imports during this time totalled US$1.29 billion. A large share of these imports comes from China. This often includes specialty cables, high-tech components, or lower-end products where Chinese manufacturers maintain a cost edge. Relying on China for certain items poses a geopolitical risk. At the same time, it offers a great chance for local manufacturers to concentrate on substituting import

Top 6 Import Sources for W&C into India (FY2024)

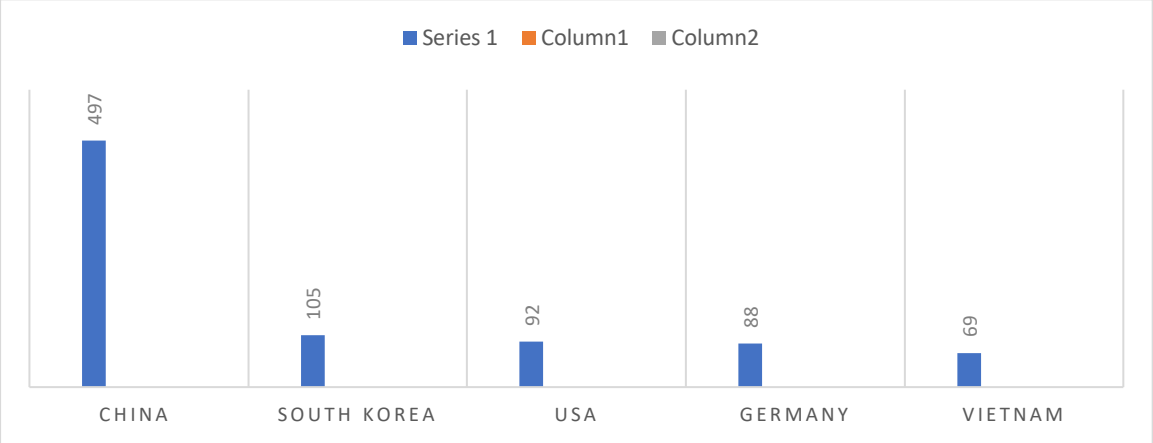


Figure 22:Top 6 Import Sources for W&C into India (FY2024)

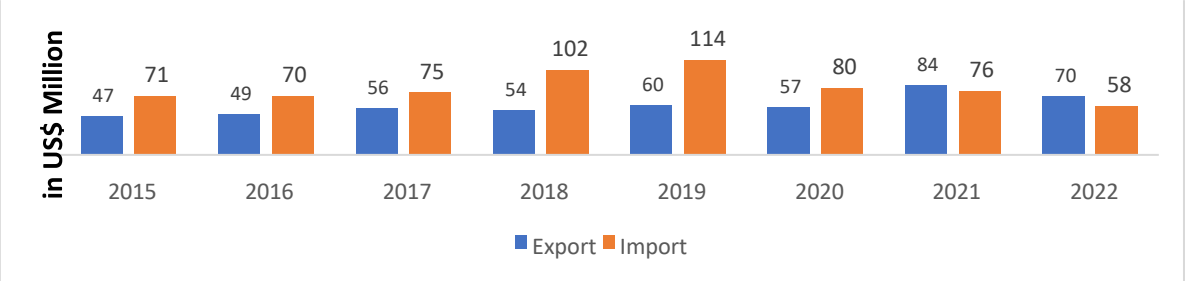


Figure 21:: Export and Import of W&C in US$ Million (CY)

Source: Trend Economy, data for calendar year

Export and Import of W&C in US$ Million (CY)

Indian W&C industry exported products valued approximately US$ 1,717 million in CY 2022. It majorly exports to USA constituting approximately 23% of exports, followed by UAE (8%) and UK (8%). As of CY 2022, RR Kabel is one of the leading exporters of wires and cables from India, in terms of value, representing an approximately 9% market share (pertaining to FY 23 revenue of RR Kabel) of the exports market from India. Approximately US$ 1,217 million worth of W&C products were imported from across the globe by India in CY 2022, wherein China accounts for approximately 34% of the imports followed by USA (10%), Germany (8%) and Korea (5%).

Indian W&C market became net export positive for the first time in CY 2019 and has been export positive since then. This shows the widespread acceptability of Indian W&C product worldwide. W&C export market has grown at a CAGR of 11% from CY 2015 to CY 2022 and is expected to grow positively in future. This growth would be driven primarily by four factors:

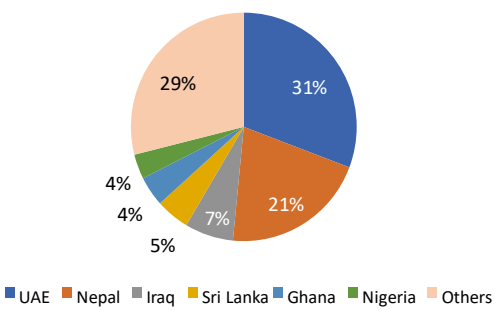
- *China plus one strategy:* In 1990s, many global manufacturing entities in geographies such as US and Europe shifted their production facilities to China owing to favourable factors of production, which made it the centre of global supply chain. But in CY 2021, when there was a surge in demand across the world post-COVID-19 pandemic, China's zero COVID policy and supply chain disruption issues made it difficult for these

manufacturing entities to meet the demand. As a result, such companies are contemplating diversifying their business and investment out of China to alternative destinations. Technopak believes that this China plus one strategy presents a great opportunity for India because of its large manufacturing base, favourable factors of production, strong business ecosystem, and incentivizing government policies, which in turn, is expected to help in growing the exports market of Indian W&C industry. Leading players like RR Kabel are well-positioned to benefit from the global shift away from China-based manufacturing to China plus one strategy resulting in a share gain for Indian manufacturers in the global market.

- *Export Incentives:* Various policy interventions by GoI to promote export of goods and services is expected to further boost the exports market for W&C in India. For instance, AAS and the EPCG Scheme are being implemented to enable duty free import of raw materials and capital goods for export production. RoDTEP scheme has also been introduced which rebates various central, state, and local duties/ taxes on exported products.
- *Technical and Regulatory Compliance:* Many stringent technical standards and compliances have been laid out by various countries for wires and cables product. For example, Europe has laid out regulatory compliances such as REACH, ROHS and CPR. RR Kabel, a leading company, has over 35 international product certifications, which has enabled them to export its products to global markets.
- *Transition towards clean and green energy:* Commitment of various countries towards achieving a cleaner and greener environment, for instance, the Paris agreement, and net zero emission have resulted in increasing adoption of renewable projects across the globe. These are expected to increase the demand for W&C in the global market, which in turn would provide a great opportunity for Indian exports

Export and Import share by countries of W&C (CY 2022)

Export Share by countries (CY 22)



Import Share by countries (CY 22)

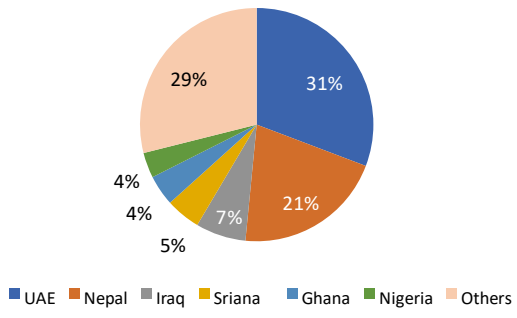


Figure 23:Export and Import share by countries of W&C (CY 2022)

Source: RR Kabel DRHP,

4.12. Key Catalysts and Headwinds

4.12.1. Catalysts:

Sustained Infrastructure Push: The government is still focusing on power T&D, railways, metros, and smart cities. This focus remains the main driver of demand.

Private Capex Revival: A rise in private spending on industrial and real estate projects would significantly boost demand across all cable segments.

Digital and Green Transitions: Non-discretionary spending on 5G fib erization, data centres, and renewable energy projects offers clear, long-term growth potential.

Stricter Standards Enforcement: The strict implementation of BIS standards will keep pushing market share from unorganized players to organized ones, which benefits compliant companies.

Export Momentum: The "China+1" global sourcing trend gives Indian manufacturers a chance to increase their presence in international markets.

4.13. Risks & Headwinds:

*Commodity Price Volatility***:** Sharp and unexpected swings in copper and aluminium prices remain the biggest risk to short-term profits and working capital management.

Execution Delays: A slowdown in government infrastructure projects or a downturn in the housing market could dampen demand growth.

Increased Competition: The arrival of large, well-funded companies could disrupt market pricing and pressure the margins of existing players.

Global Economic Slowdown: A major global recession could hurt export growth and demand for OFC, which is priced on a global scale.

Technological Disruption: While it presents an opportunity, the rapid development of technology, such as new transmission methods and wireless options, requires ongoing investment in research and development to stay competitive.

chapter 5: Operational and Financial Benchmarking

5.1. Review of financial performance

A detailed financial analysis of the five leading companies from FY21 to FY25 reveals clear patterns in growth, profitability, and operational efficiency. The following tables and analyses offer a detailed, data-based comparison of their performance over this five-year period.

5.2 Revenue from Operations

Revenue from operations is the top line parameter for a company's financials. Havells, Polycab and KEI Industries were the top three companies in terms of revenue generation in Fiscal 2023. RR Kabel is the fastest growing consumer electrical company among the peers in India, growing at a CAGR of 43.4% between Fiscal 2021 and Fiscal 2023.

Revenue from Operations in Fiscal Year (₹ Crore)

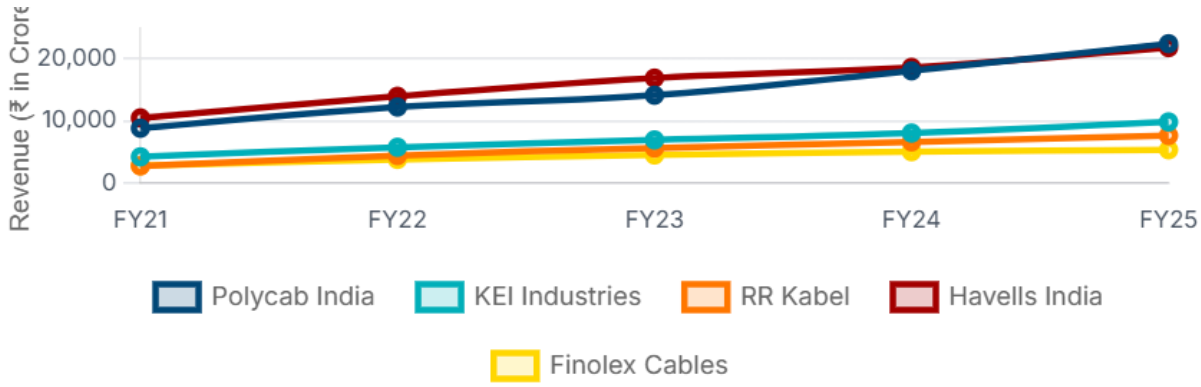


Figure 24:Revenue from Operations in Fiscal Year (₹ Crore)

Polycab India has established a dominant lead in market scale, while challengers like RR Kabel and KEI Industries demonstrate the most aggressive growth trajectories over the last five years.

Revenue from Operations in Fiscal Year (₹ Crore)

Table 10:Revenue from Operations in Fiscal Year (₹ Crore)

Company	2021	2022	2023	CAGR 2021-23
RR Kabel	2,724	4,386	5,599	43.4%
Finolex Cables	2,768	3,768	4,481	27.2%
Polycab	8,792	12,204	14,108	26.7%
KEI Ind	4,182	5,727	6,912	28.6%
Havells	10,457	13,938	16,911	27.2%
Bajaj Electricals	4,585	4,813	5,429	8.8%
Crompton	4,804	5,394	6,870	19.6%
V Guard	2,721	3,498	4,126	23.1%

Source: RR Kabel DRHP,

5.3. EBITDA and EBITDA margin

EBITDA (₹ Crore) and EBITDA margin

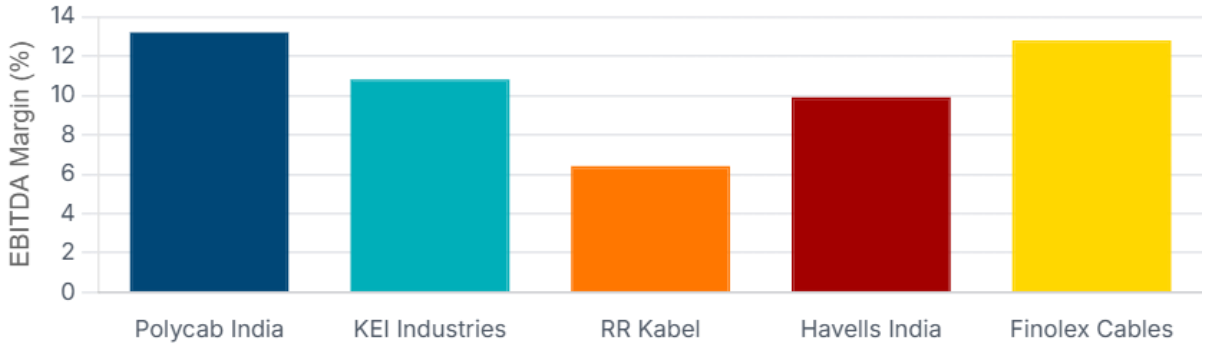


Figure 25:BITDA (₹ Crore) and EBITDA margin

EBITDA margins reveal a company's operational efficiency and pricing power. Polycab's scale and KEI's strategic retail pivot yield strong, stable margins, while RR Kabel shows more volatility due to commodity exposure.

EBITDA (₹ Crore) and EBITDA margin (%) profiles in Fiscal Year

Table 11:EBITDA (₹ Crore) and EBITDA margin (%) profiles in Fiscal Year

Company	2021		2022		2023		Fiscal 2021-2023
	EBITDA	*Margin*	*EBITDA*	*Margin*	*EBITDA*	*Margin*	CAGR
RR Kabel	252	9.2%	350	7.9%	357	6.3%	19.0%
Finolex Cables	447	15.7%	500	13.0%	627	13.6%	18.4%
Polycab	1,230	13.8%	1,355	11.0%	1,985	13.9%	27.0%
KEI Ind	476	11.3%	603	10.5%	734	10.6%	24.2%
Havells	1,759	16.5%	1,908	13.5%	1,777	10.4%	0.5%
Bajaj Electricals	423	9.1%	292	6.0%	434	7.9%	1.2%
Crompton	796	16.3%	816	14.9%	837	12.1%	2.5%
V Guard	333	12.1%	351	10.0%	336	8.1%	0.5%

Source: RR Kabel DRHP,

Note: EBITDA Margin = EBITDA/ Total Income Note: EBITDA = Profit Before Tax + Depreciation & Amortization + Finance Costs – Exceptional Items (if any) Note: CAGR shown on basis of EBITDA value.

EBITDA and EBITDA margins is largely used to compare the profitability of the companies against competitors. It is also used to standardize the business performance against the industry averages. Polycab, KEI Industries and RR Kabel have registered the highest EBITDA CAGR of 27.0%, 24.2% and 19.0% respectively for the period Fiscal 2021 to Fiscal 2023.

5.4. PAT and PAT margin

The trend in PAT margins is similar to EBITDA. Polycab consistently provides the highest net profitability. The profit after tax (PAT) and PAT margins are used to assess if a company's business is profitable after meeting the operating and overhead costs. The PAT margin is under pressure across all players in the industry. Finolex Cables registered the highest CAGR of 35.1% for PAT for the time period Fiscal 2021 to Fiscal 2023. RR Kabel registered a PAT CAGR of 18.7% during the same period.

Table 12:. PAT and PAT margin

Company	Metric	FY21	FY22	FY23	FY24	FY25
Polycab India	ROE (%)	17.0	15.0	18.0	20.0	19.8
	ROCE (%)	22.0	20.0	24.0	27.0	26.0
KEI Industries	ROE (%)	16.0	18.0	18.5	18.5	12.1
	ROCE (%)	21.0	24.0	26.2	26.3	17.2
RR Kabel	ROE (%)	16.0	16.0	13.4	16.3	14.5
	ROCE (%)	18.0	18.0	14.8	19.8	19.8
Havells India	ROE (%)	19.0	18.0	15.0	17.1	17.7
	ROCE (%)	26.0	25.0	21.0	24.1	24.8
Finolex Cables	ROE (%)	10.0	11.0	11.5	13.2	N/A
	ROCE (%)	13.0	14.0	14.9	17.5	N/A

Top Performers: Polycab India and Havells India consistently show high ROE and ROCE figures. This indicates strong profitability and efficient use of capital. Polycab's ROCE is especially impressive, reaching 27.0% in FY24.

Notable Downturn: KEI Industries displayed strong and stable returns through FY24. However, it faced a significant drop in both ROE and ROCE in FY25.

Steady Improver: Finolex Cables shows a clear, positive trend in improving its returns year-over-year. Still, its absolute figures are lower than those of the industry leaders.

Variable Performance: RR Kabel maintains moderate returns. Its performance fluctuates more compared to its peers, showing a noticeable dip in FY23 followed by a recovery.

5.4.Return on Equity

Return on equity (ROE) is a function of profit after tax and shareholder's equity. A sustainable business and increasing ROE over time can indicate a good value generation ability for the shareholders. A negative ROE indicates negative profit earned by the business. Polycab had the highest ROE among industry peers in Fiscal 2023.

Table 13:Return on Equity

Company	Metric	FY21	FY22	FY23	FY24	FY25	Key Observations
Polycab India	ROE (%)	17.0	15.0	18.0	20.0	19.8	Consistently strong ROE, peaking in FY24.
	ROCE (%)	22.0	20.0	24.0	27.0	26.0	Excellent and improving capital efficiency.
KEI Industries	ROE (%)	16.0	18.0	18.5	18.5	12.1	Stable performance until a significant drop in FY25.
	ROCE (%)	21.0	24.0	26.2	26.3	17.2	Strong ROCE that mirrored the sharp decline in FY25.
RR Kabel	ROE (%)	16.0	16.0	13.4	16.3	14.5	Moderate and somewhat fluctuating returns.
	ROCE (%)	18.0	18.0	14.8	19.8	19.8	Steady improvement in capital use after a dip in FY23.
Havells India	ROE (%)	19.0	18.0	15.0	17.1	17.7	Solid ROE with a recovery trend after the FY23 dip.
	ROCE (%)	26.0	25.0	21.0	24.1	24.8	High ROCE, showing efficient capital management.
Finolex Cables	ROE (%)	10.0	11.0	11.5	13.2	N/A	Steady, gradual improvement in shareholder returns.
	ROCE (%)	13.0	14.0	14.9	17.5	N/A	Consistently improving efficiency year-over-year.

Top Performers: Polycab India and Havells India consistently show the highest returns for both ROE and ROCE. This indicates efficient management and strong profitability. Polycab, in particular, shows impressive growth in its ROCE.
Steady Performer: KEI Industries was a strong and steady performer until the significant downturn in FY25. This situation deserves a closer look.
Improving Efficiency: Finolex Cables shows a clear and positive trend in its returns and capital efficiency each year, even if its absolute numbers are lower than those of the leaders.

Moderate Performer: RR Kabel maintains decent returns but with more variability compared to its peers.

5.5.Working Capital Cycle (Days)

Table 14:Working Capital Cycle

Company	FY21	FY22	FY23	FY24	FY25
Polycab India	60	55	52	50	49
KEI Industries	95	90	88	85	82
RR Kabel	75	70	68	64	56
Havells India	65	62	60	58	55
Finolex Cables	100	98	95	92	N/A

In short, the data shows that these companies faced a tough five-year period across all measured metrics. Finolex and KEI began as the leaders, but each of them saw a decline. Polycab India, although it started at the bottom, seems to have handled the downward pressure the best, having the slowest rate of decrease.

5.6 Company-wise Mini Takeaways

5.6.1. Polycab India

Growth: The company shows strong, consistent growth at 26.2% over the last five years. This growth comes from gaining market share in W&C and expanding FMEG.

Margins: The company boasts industry-leading EBITDA margins of 13 to 14%. This reflects strong brand loyalty and pricing power.

WC Levers: The company has excellent working capital management, averaging 49 days. This is backed by a solid distribution network and channel financing.

Capex: There is an aggressive capital expenditure plan of ₹9.6 billion for FY25 to meet future demand.

Positioning: The company is a dominant leader in W&C and is successfully branching out into a full range of electrical goods.

5.6.2. KEI Industries

Growth: The company has high revenue growth of 23.6% over a five-year period. It focuses on expanding its retail and export segments.

Margins: The margins are strong, supported by a valuable product mix. The company holds a leading position in the EHV cables segment.

WC Levers: There is a longer working capital cycle due to significant involvement in the institutional and EPC business, which has longer payment terms.

Capex: The company is investing heavily in new capacity, especially at the Sanand plant, to promote future growth.

Positioning: It is a solid number two in the W&C space with a balanced B2B and B2C portfolio and specialized EHV capabilities.

5.6.3. RR Kabel

Growth: The company has the highest revenue growth among its peers at 29.3% over the last five years. This growth comes from strong volume increases in both domestic and export markets.

Margins: Although margins are currently lower, they are expected to improve as the FMEG business grows and reaches breakeven.

WC Levers: The company's management of working capital is exceptional at 56 days, which provides a significant competitive edge.

Capex: The company announced a new capital expenditure of ₹1,200 crore, which will be completed by FY28, aimed at expanding cable production capacity.

Positioning: This is a fast-growing company with a strong focus on B2C, using its brand to enter the FMEG market. It is the largest exporter of wires and cables in India.

5.6.4. Havells India

Growth: The company has steady growth at 20.0% over five years. This is driven by its diverse portfolio and strong brand appeal.

Margins: The margins are stable and healthy, though the W&C segment's margins are a bit lower than those of pure-play leaders.

WC Levers: The working capital cycle is efficient, benefiting from a large and well-established distribution network.

Capex: The company continually invests in building its brand and improving technical skills to keep its premium position.

Positioning: The company is a diverse FMEG giant where W&C is an important but complementary business area.

5.6.5. Finolex Cables

Growth: There is moderate revenue growth, and the company has a solid legacy brand in electrical and communication cables.

Margins: The company maintains consistently high profit margins, supported by its backward integration into copper rod manufacturing.

WC Levers: The working capital cycle is longer than that of its peers, suggesting room for operational improvement.

Capex: The company plans to expand production capacity for its entire product range.

Positioning: This is an established, debt-free company with a strong brand legacy that focuses on its cable business.

Chapter 6 Conclusion & Recommendations

6.1. Conclusion

The Indian wires and cables industry is at a crucial point and ready for a period of fast growth. A large increase in public and private spending on infrastructure, a boom in housing, and the significant impact of green and digital economies create strong demand for the foreseeable future.

The competitive environment increasingly Favors organized companies that can use their scale, brand reputation, and wide distribution networks to gain market share from smaller, unorganized players. Success in this changing landscape will depend on a company's ability to achieve two goals: maintaining operational excellence in high-volume, standard markets while also driving technological innovation in high-profit, specialized areas like EHV cables, optical fibre, and tailored solutions for electric vehicles and renewable energy.

While challenges like commodity price fluctuations and growing competition continue, the overall outlook remains positive. The companies that will stand out as leaders will be those that not only meet current demand but also anticipate the future needs of a developing India. Investing in technology, building brand strength, and managing capital efficiently are now essential for navigating the competitive landscape and providing long-term value to stakeholders.

6.2. Recommendations

To take advantage of the identified growth drivers and reduce inherent risks, companies in the Indian Wires and Cables sector should consider these strategic initiatives:

Develop Integrated "Green Energy" Cabling Solutions. The goal of 500 GW of renewable capacity by 2030 presents a significant opportunity. Instead of just offering standard solar and wind cables, companies should shift to providing integrated "grid-to-generation" solutions. This involves creating bundled packages of high-performance E-Beam cross-linked cables, specialized HV/EHV transmission lines for solar parks, and IoT-enabled smart cables that allow real-time monitoring of energy loss and performance. Building a distinct "Green Cable" brand, certified for sustainability and efficiency, would attract clients focused on ESG and command a premium.

Establish a "Cabling-as-a-Service" (CaaS) Model for Data Centres. The expected increase in data centre capacity to 9 GW by 2030 requires a massive and reliable cabling infrastructure. Leading companies should pilot a CaaS model to offer hyperscale data centre operators a complete, managed service for their cabling needs. This would cover design, installation, lifecycle management, and technology upgrades for both power and high-speed optical fibre cables on a long-term subscription basis. This approach shifts the business model from one-time sales to a recurring revenue stream, fostering stronger client partnerships.

Create an AI-Powered Ecosystem for Electricians and Contractors. Electricians are key influencers in the B2C and small B2B segments, so companies should go beyond basic loyalty programs. A new digital platform could provide AI-driven project management tools, wiring layout suggestions based on architectural plans, and direct access to technical support. Integrating this platform with an Augmented Reality (AR) app would help homeowners visualize switch placements and lighting solutions, creating a powerful data-rich ecosystem that strengthens brand loyalty and offers valuable insights into end-user behaviour.

Pioneer Specialized Cabling for the EV and Advanced Mobility Sector. The goal of installing 2 million EV charging stations by 2030 opens up a high-tech market. Companies should invest in R&D to become leaders in specialized EV cabling. This includes creating ultra-durable, high-current fast-charging cables, lightweight and heat-resistant in-vehicle wiring harnesses, and smart cables with integrated sensors for battery management systems. Forming early partnerships with automotive OEMs and charging infrastructure providers will be crucial to capturing this emerging market.